



Experimental Liquidus Studies of the Pb-Fe-Si-O System in Air

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Abstract Phase equilibria of the Pb-Fe-Si-O system have been investigated at 958–1913 K (685–1640 °C) for oxide liquid in equilibrium with air and solid oxide phases: (a) quartz, tridymite or cristobalite SiO_2 ; (b) hematite Fe_2O_3 ; (c) spinel $\text{Fe}_3\text{O}_{4+x}$; (d) complex lead-iron silicates (melanotekite $\text{PFS} = \text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$, barysilite $\text{P}_{9-x}\text{F}_x\text{S}_6 = (9-x)\text{PbO} \cdot x\text{FeO}_{1+y} \cdot 6\text{SiO}_2$, “ P_5FS ” = $5\text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$, and “ P_6FS ” = $6\text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$); (e) lead silicates (alamosite $\text{PS} = \text{PbO} \cdot \text{SiO}_2$, $\text{P}_2\text{S} = 2\text{PbO} \cdot \text{SiO}_2$, $\text{P}_{11}\text{S}_3 = 11\text{PbO} \cdot 3\text{SiO}_2$, $\text{P}_5\text{S} = 5\text{PbO} \cdot \text{SiO}_2$); (f) lead ferrites (magnetoplumbite $\text{PbO} \cdot 12\text{FeO}_{1.5}$, plumboferrite $\text{PbO} \cdot (5+x)\text{FeO}_{1.5}$, 1:1 lead ferrite $\text{PbO} \cdot (1 \pm x)\text{FeO}_{1.5}$); and (g) lead oxide (PbO , massicot). High-temperature equilibration on primary phase or inert metal (platinum, gold) substrates, followed by quenching and direct measurement of Pb, Fe and Si concentrations in the phases with the electron probe x-ray microanalysis (EPMA) has been used to accurately characterize the system in equilibrium with air. All results are projected onto the PbO –“ $\text{FeO}_{1.5}$ ”– SiO_2 plane for presentation purposes. The present study is the first systematic characterization of liquidus over a wide range of compositions in this system in equilibrium with air.

Keywords iron · lead · phase diagrams · silica · slags

1 Introduction

The chemical compositions of slags in typical lead smelting operations can be represented by the Pb-Zn-Fe-Cu-Ca-Si-Al-Mg-O system. Accurate data on slag-solid-metal phase equilibria, activities of lead species in the slag, and partitioning of elements between gas, slag, metal and solid phases are essential to support the further optimization and development of complex lead smelting, refining and recycling processes.

The present experimental studies of gas-slag-solid oxide phase equilibria in the Pb-Fe-Si-O “low-order” system is an important part of an integrated thermodynamic modeling and experimental research program for the above multicomponent system. The earlier studies of the lead-containing slag systems^[1–11] were focused on the multicomponent Pb-Zn-Fe-Ca-Si-O equilibria in air. Glasser^[12] examined a subsolidus isothermal section of the PbO –“ Fe_2O_3 ”– SiO_2 system in air at 923 K (650 °C). Some information on the liquidus surface in air can also be derived from the work by Nitta (1989).^[13] Some preliminary studies on the solid–liquid equilibria in the PbO –“ Fe_2O_3 ”– SiO_2 system in air have been performed using the equilibration/quenching/microanalysis technique.^[14] The information available at that time was used for the development of the thermodynamic database^[15] for the system Pb-Zn-Fe-Ca-Si-O phase equilibria based on the results at different conditions (including liquidus in air^[14] and in equilibrium with metallic iron^[16]), using FactSage.^[15,17–19] Accurate thermodynamic predictions require experimental data on all low-order systems (binary and ternary) and all phases that can be formed. Significant gaps in experimental data in the Pb-Zn-Fe-Cu-Si-Ca-Al-Mg-O system have been identified to be essential for revision of the current thermodynamic database.^[15,19] Several issues have been

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identified in the preliminary dataset,^[14] that required further experimental work. The study on the liquidus of the PbO–“FeO”–SiO₂ system in equilibrium with metallic Pb has been completed recently.^[20] To evaluate the effect of oxygen potential on phase equilibria at oxidizing conditions, the current study of the liquidus in the PbO–“FeO_{1.5}”–SiO₂ system in air is conducted.

According to the current public FactSage 6.2 database,^[15,17–19] most of the liquidus surface in the PbO–“FeO_{1.5}”–SiO₂ system in equilibrium with air belongs to the quartz, tridymite, cristobalite (SiO₂), hematite (Fe₂O₃), spinel (Fe₃O_{4+x}), and magnetoplumbite (“PbFe₁₀O₁₆”) primary phase fields. Liquid immiscibility was expected at very high temperatures (> 1873 K (1600 °C)) over the cristobalite liquidus field. Other predicted primary phase fields were melanotekite (PbO·FeO_{1.5}·SiO₂), 5PbO·FeO_{1.5}·SiO₂, lead silicates (alamosite PbSiO₃, lead orthosilicate Pb₂SiO₄, and “4-lead silicate Pb₄SiO₆”), lead ferrites (1:1 lead ferrite PbO·FeO_{1.5}, plumboferrite “PbO·4FeO_{1.5}”), and lead oxide (massicot, PbO). However, no experimental studies reporting the exact extents of these phase fields were found in literature. The current study focuses on the equilibria of the slag-(solid oxide)-air phases over wide range of temperatures (938–1743 K (685–1470 °C)) and compositions.

2 Experimental Technique and Procedure

2.1 Experimental Procedure and Sample Examination

The experimental procedure and apparatus have been described in detail in previous publications by PYROSEARCH.^[11,20,21] Initial mixtures were made by mixing high-purity powders of Fe₂O₃ (99.945 wt.% purity), PbO (99.9 wt.% purity), SiO₂ (99.9 wt.% purity), supplied by Alfa Aesar, MA, USA.

Four types of substrates were used for equilibration, depending on the conditions:

1. Silica open crucibles or ampoules with drilled holes (to ensure contact with air but avoid excessive evaporation of PbO).
2. Hematite (Fe₂O₃) baskets, prepared by heating a folded iron foil (99.5 wt.% purity Fe, 0.1 mm thickness, supplied by Sigma Aldrich, MO, USA) in air at 1473 K (1200 °C) for 2–3 h (the stoichiometry of Fe₂O₃ was checked gravimetrically), were used for the experiments where hematite was one of the primary phases.

3. Platinum foil envelopes (> 99.9% purity, 0.05 mm thickness, provided by Johnson Matthew, Australia) were used for equilibration of slags with melanotekite (PbO·FeO_{1.5}·SiO₂), magnetoplumbite (PbO·12FeO_{1.5}), and sometimes hematite (Fe₂O₃) and spinel (Fe₃O_{4+x}) at temperatures above 1073 K (800 °C).
4. Gold foil baskets (99.99% purity, 0.127 mm thickness, provided by Sigma Aldrich, Australia) were used for equilibration of PbO-rich slags with various primary solid phases at or below 1073 K (800 °C), because platinum reacts at such conditions forming lead platinates PbPt₂O₄ and Pb₂PtO₄.^[22]

The technique of equilibration experiments and parameters of electron probe x-ray microanalysis (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan) were similar to Ref 20. The sample was suspended in the hot zone of the furnace by Kanthal support wire (0.7 or 1-mm diameter) at temperatures below ~ 1753 K (1480 °C); at higher temperatures, Pt or Pt–Rh wires were used. Samples were pre-melted for various times (0.5–19 h, see Table 2) at 20–200 K above the equilibration temperature, to form a homogeneous slag. This was followed by equilibration at the final target temperature for the required time (from 0.5 to 1 h in tridymite and spinel primary phase fields at high temperatures to 3–5 days for quartz primary phase field at low temperature, see Table 2), to allow solid phases to crystallize from the melts. For some experiments where the crystallization from the viscous slag was slow, in particular, for the lead and lead-iron silicates liquidus below 1073 K (800 °C), an extra pre-crystallization stage was introduced between the premelt and the final equilibration. To avoid undercooling, the sample was kept below the final temperature for several hours or days, up to 4 months for the PbSiO₃ primary phase field.

Hematite (Fe₂O₃), quartz (SiO₂), wollastonite (CaSiO₃) (supplied by Charles M. Taylor Co., Stanford, CA, USA), and K-456 lead silicate glass (71 wt. pct. PbO, supplied by NIST, Wash. D.C., USA) standards were used for calibration of EPMA. Based on the previous work,^[20] where fayalite (Fe₂SiO₄) and lead silicates (PbSiO₃ and Pb₂SiO₄) served as additional stoichiometric standards in the Fe–Si–O and Pb–Si–O systems, subsequent correction^[23] was applied to the compositions of all phases obtained after the JEOL ZAF correction:

$$x_{\text{SiO}_2}^{\text{corrected}} = x_{\text{SiO}_2} - x_{\text{SiO}_2} x_{\text{PbO}} \\ \left(-0.0129 - 0.0206 x_{\text{SiO}_2} + 0.421 x_{\text{SiO}_2}^2 - 0.4066 x_{\text{SiO}_2}^3 \right) \\ - 0.0637 x_{\text{FeO}} x_{\text{SiO}_2} (1 - x_{\text{SiO}_2}); \quad (\text{Eq } 1)$$

$$x_{\text{PbO}}^{\text{corrected}} = x_{\text{PbO}} \frac{1 - x_{\text{SiO}_2}^{\text{corrected}}}{1 - x_{\text{SiO}_2}}; \quad (\text{Eq 2})$$

$$x_{\text{FeO}}^{\text{corrected}} = x_{\text{FeO}} \frac{1 - x_{\text{SiO}_2}^{\text{corrected}}}{1 - x_{\text{SiO}_2}}. \quad (\text{Eq 3})$$

Kawasaki et al.^[24] reported that secondary x-ray fluorescence can result in significant overestimation of the “FeO_{1.5}” concentration measured by EPMA in cristobalite within 50 micron from the high-iron phase: values up to 1.1 mol.% “FeO_{1.5}” were observed from EPMA whereas the solubility of iron oxide in cristobalite in equilibrium with hematite and magnetite and did not exceed 0.28 mol.%. A special experiment undertaken in the present study with 50–200 micron size grains of quartz pressed together with Fe₂O₃ powder (unreacted couple) confirmed that apparent 0.7 mol.% “FeO_{1.5}” is observed in pure quartz grains due to the secondary x-ray fluorescence. The following correction for secondary x-ray fluorescence therefore has been developed and applied in the present study to the compositions of tridymite and cristobalite measured by EPMA:

$$x(\text{FeO}_{1.5} \text{ in SiO}_2)^{\text{corr}} = x(\text{FeO}_{1.5} \text{ in SiO}_2)^{\text{initial}} - 0.007 \text{ wt. fraction}(\text{FeO}_{1.5} \text{ in slag}). \quad (\text{Eq 4})$$

The ratios of the different oxidation states of iron cations were not measured with EPMA, and the iron oxide concentrations were recalculated as FeO_{1.5} for presentation purposes. According to FactSage predictions, the ferrous (FeO) content in slags are very low across the whole range of concentrations except very high temperatures (spinel primary phase field and two liquid miscibility gap over the cristobalite primary phase field).

Significant quenching difficulties were observed for the high-iron oxide slags in the spinel and hematite primary phase fields, where the areas of glassy homogeneous slag phase were limited to locations at the surfaces, directly contacting the quenching medium, and the upper limit of temperatures could not exceed 1673 K (1400 °C). Overall, the glass-forming ability of the PbO–“FeO_{1.5}”–SiO₂ slags in air has been observed to be much better than the PbO–“FeO”–SiO₂ slags in equilibrium with Pb metal, possibly due to the fact that Fe³⁺ ion acts as a moderate glass-former rather than Fe²⁺ that promotes highly fluid slags.

The absence of evaporation loss of Pb species during EPMA measurement of a single point up to 1 min^[20] was confirmed in the present study. The accuracy of the EPMA measurements was estimated to be within 1 wt.%, and the typical detection limit of minor components was estimated to be about 0.01 wt.%.^[10]

2.2 Confirmation of Achievement of Equilibrium

To ensure the achievement of equilibrium in the samples, the four-point test approach^[21,25] was used. Dedicated series of experiments were carried out for these purposes, resulted in the experimental design features described below.

Lead silicate master-slag (the eutectic at 40 mol.% PbO melting < 1013 K (740 °C)) was also used to reduce Pb/PbO vaporization during the initial heating stage, to suppress foaming of the slag, and to enhance homogeneity of the sample. Consequently, all initial mixtures were prepared from (SiO₂ + Fe₂O₃ + master-slag) or (Pb₃O₄ + Fe₂O₃ + master-slag) powders. Pb₃O₄ powder, prepared from PbO heated in air at 723 K (450 °C) for 24 h, was used instead of PbO to protect Pt and Au substrates from alloy formation with Pb metal and destruction that could happen locally due to reducing conditions.

Fe₂O₃ powder particles have been found to stay in finely dispersed state in the samples at hematite, melanotekite, and magnetoplumbite primary phases at T < 1273 K (1000 °C), creating unfavorable conditions for EPMA measurements of slag compositions, and potentially affecting quenching. Therefore, the excess of Fe₂O₃ in the initial mixture was carefully avoided, so that the initial premelt was able to convert the sample to fully liquid state, from which larger crystals of the primary phase would precipitate at final equilibration temperature.

A major source of incomplete achievement of equilibrium in the present study was found to be a failure of some phases to nucleate from viscous slag at low temperatures. In particular, the melanotekite (PbFeSiO_{4.5})-hematite (Fe₂O₃), melanotekite (PbFeSiO_{4.5})-quartz (SiO₂), and melanotekite (PbFeSiO_{4.5})-alamosite (PbSiO₃) boundaries required long experimental time and careful selection of preheat and precrystallization heat treatments. This selection was based on estimations of slag viscosity, lead species evaporation rate, and possible uncertainties due to specific slow reactions to achieve equilibrium and quenching issues. In particular, proximity to quartz and alamosite primary phase fields, corresponding to very viscous slags, required long equilibration times (up to 1 week), sometimes also preceded by several weeks of precrystallization at subsolidus temperatures, since the nucleation rate of quartz and alamosite has been found to be the rate-limiting step for achievement of equilibrium. Selection of preheat to higher temperature relative to the final equilibration temperature was particularly necessary for samples with hematite or melanotekite at low temperatures. Without a significant preheat, those solids formed very fine crystals distributed throughout the slag as dust, which makes difficult to measure both solid and liquid

compositions (see sample 102 in Table 2). Preheat to 100–200 K above the final temperature allows to dissolve all or most of these fine crystals, after that the new crystals grow on much fewer nucleation sites and therefore reach larger size. In other cases, when the vapor pressure of lead species was high, preheat was avoided, since it would further increase evaporation loss. Instead, the initial bulk composition was planned with small excess of PbO, so that the crystals (hematite, spinel, tridymite, cristobalite, magnetoplumbite) can be almost dissolved at the beginning of equilibration, and then gradually grow due to evaporation of PbO and change of bulk composition towards the $\text{FeO}_{1.5}$ - SiO_2 direction.

3 Results and Discussion

Examples of micrographs of quenched PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 samples are given in Fig. 1(a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l), (m), (n), (o), and (p). All phases observed in the system in the present study are listed in Table 1. Note that the abbreviations accepted in the present study are P = PbO, S = SiO_2 , F = FeO or $\text{FeO}_{1.5}$ (not Fe_2O_3), i.e. the iron atoms are counted regardless of their oxidation state, unambiguously reporting the concentrations in terms of the metal cation ratios and projecting the concentrations onto the PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 plane.

In the previous study,^[20] some phases not included into the FactSage database^[15,17–19] were discovered: barysilite “ P_8FS_6 ” = $8\text{PbO} \cdot \text{FeO} \cdot 6\text{SiO}_2$ and “ P_6FS ” = $6\text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$. Along the PbO - SiO_2 join, $\text{Pb}_{11}\text{Si}_3\text{O}_{17}$ and Pb_5SiO_7 ^[26,27] were found to be stable instead of Pb_4SiO_6 .^[15] In the Pb - Fe - O system, $\text{Pb}_{1+x}\text{Fe}_{12-x}\text{O}_{19-x}$ (magnetoplumbite^[28]) and $\text{PbFe}_{5+x}\text{O}_{8.5+1.5x}$ (plumboferrite) were found to be stable instead of $\text{PbFe}_{10}\text{O}_{16}$ and PbFe_4O_7 .^[15] In a number of samples in air, barysilite ($\text{P}_{9-x}\text{F}_x\text{S}_6$) phase was also observed; unlike $8\text{PbO} \cdot \text{FeO} \cdot 6\text{SiO}_2$ that exists at reducing conditions, it is shifted towards the $\text{Pb}_9\text{Si}_6\text{O}_{21}$ ($\text{Pb}_3\text{Si}_2\text{O}_7$) end-member that can be stable only at subsolidus conditions.^[26] The oxidation state of iron in the observed $\text{Pb}_{9-x}\text{Fe}_x\text{Si}_6\text{O}_{21+y}$ barysilite was not determined, with possible variation of y from 0 (Fe^{2+}) to $x/2$ (Fe^{3+}); $0 < x < 1$.

The initial data from EPMA (normalized wt.% of oxide components PbO, “ Fe_2O_3 ” and SiO_2 and their initial totals), as well as the results of experiments after correction according to Eq 1, 2 and 3 are reported in Table 2. The phase diagram constructed based on the present results is given in Fig. 2(a) and (b). The pseudobinary diagram PbO -“ $\text{FeO}_{1.5}$ ”, updated at high temperature part after the previous study,^[28] is shown in Fig. 2c. The diagram in wt.% of components (without the experimental points) is given in Fig. 3(a) and (b). The estimated liquidus invariant points involving the observed phases are listed in Table 3.

Fig. 1 Backscattered scanning electron micrographs of quenched slag in the PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 system in equilibrium with one or two crystalline phases and air, $p(\text{O}_2) = 0.21$ atm: (a) Liquid + PF ($\text{PbFeO}_{2.5}$) + P_6FS ($\text{Pb}_6\text{FeSiO}_{9.5}$); (b) Liquid + P_2S (Pb_2SiO_4) + PFS (melanotekite, $\text{PbFeSiO}_{4.5}$); (c) Liquid + PFS (melanotekite, $\text{PbFeSiO}_{4.5}$); (d) Liquid + PbO (massicot) + PF ($\text{PbFeO}_{2.5}$); (e) Liquid + P_2S (Pb_2SiO_4) + P_{11}S_3 ($\text{Pb}_{11}\text{Si}_3\text{O}_{17}$); (f) Liquid + tridymite (SiO_2) + hematite (Fe_2O_3); (g) Liquid + PFS (melanotekite, $\text{PbFeSiO}_{4.5}$); (h) Liquid + hematite (Fe_2O_3) + magnetoplumbite (PF_{12}); (i) Liquid + PbO (massicot) + P_6FS ($\text{Pb}_6\text{FeSiO}_{9.5}$); (j) Liquid + spinel ($\text{Fe}_3\text{O}_{4+x}$) + tridymite (SiO_2); (k) PbSiO_3 + Pb_2SiO_4 + barysilite ($\text{Pb}_{9-x}\text{Fe}_x\text{Si}_6\text{O}_{21+y}$); (l) Liquid + PbSiO_3 + quartz (SiO_2) + PFS (melanotekite, $\text{PbFeSiO}_{4.5}$); (m) Liquid + cristobalite (SiO_2); (n) Liquid + P_5FS ($\text{Pb}_5\text{FeSiO}_{8.5}$); (o) Liquid + Pb_5SiO_7 ; (p) Liquid + plumboferrite (PF_5) + magnetoplumbite (PF_{12}); (q) 2 liquids + cristobalite (SiO_2)

The liquidus in quartz, tridymite and cristobalite (SiO_2) primary phase fields has been determined over a wide range of temperatures, 993–1743 K (720–1470 °C) and $\text{Fe}/(\text{Fe} + \text{Pb})$ molar ratios in slag (from 0 at the binary PbO - SiO_2 system^[26] to ~ 0.8). The quartz–tridymite and tridymite–cristobalite transitions were not determined directly and accepted from FactSage database^[15,17–19] as 1144 K (867 °C) and 1742 K (1465 °C), respectively. A range of liquid immiscibility is known to exist in the “ $\text{FeO}_{1.5}$ ”- SiO_2 system^[32,33] and it is expected to extend over a part of the ternary PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 system but not reach the binary PbO - SiO_2 system.^[26] This area involves extreme experimental difficulties due to poor quenching from high temperatures; only a few successful experiments have been obtained in the present study. The area of two immiscible liquids does not extend more than 7 mol.% PbO, which is much less than predicted by previous FactSage model.^[15,17–19] This may indicate existence of charge compensation effect that involves stabilization of tetrahedral Fe^{3+} by the large Pb^{2+} ions.

All area of hematite (Fe_2O_3) has been covered by the current investigation. Its boundary with spinel ($\text{Fe}_3\text{O}_{4+x}$), where $x \sim 0.11$ at the transition temperature 1664 K (1391 °C) in air, $p(\text{O}_2) = 0.21$ atm, according to public FactSage database.^[15,17–19] Spinel was experimentally observed at 1673 K (1400 °C) and above, and hematite—at 1663 K (1390 °C) and below, in agreement with the predictions. Several experiments have been dedicated to determine the slag-hematite (Fe_2O_3)-magnetoplumbite ($\text{PF}_{12} = \text{PbFe}_{12}\text{O}_{19}$) invariant reaction in the silica-free PbO -“ Fe_2O_3 ” system, which has been bracketed between 1593 K (1320 °C) and 1613 K (1340 °C).

The melanotekite ($\text{PFS} = \text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$) has been found to form through the peritectic reaction Liquid + hematite (Fe_2O_3) $\rightarrow$ melanotekite (PFS) estimated to occur at 1203 ± 5 K (930 ± 5 °C). The melanotekite primary phase field is assumed to be limited by seven ternary invariant points at fixed $p(\text{O}_2) = 0.21$ atm:

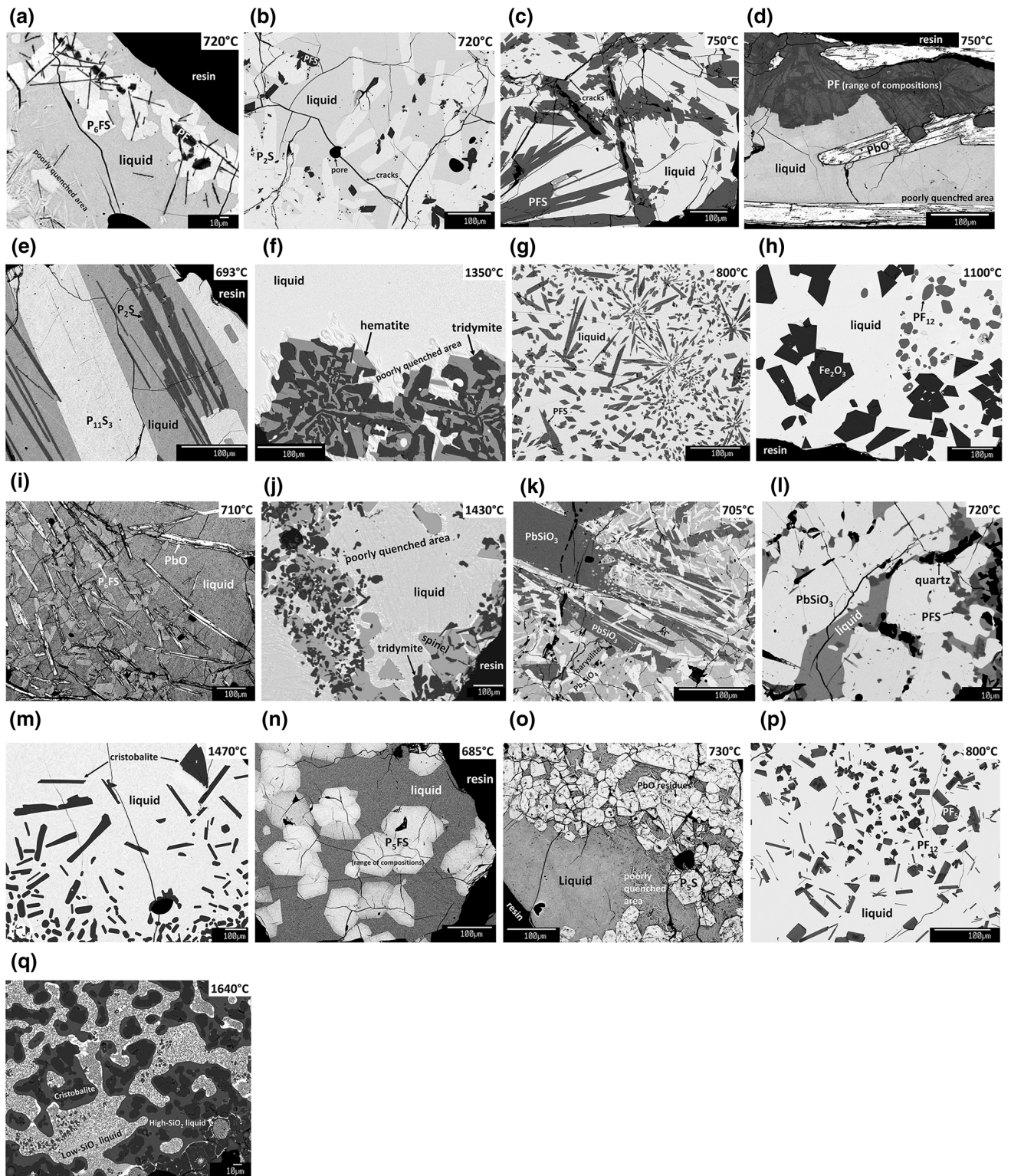


Table 1 Composition ranges of solid primary phases at the PbO–FeO_{1.5}–SiO₂ liquidus in air

Phase name	Formula	Abbrev.	Current FactSage	Present study	No. of points	Composition range
Cristobalite	SiO ₂		Yes	Yes	15	< 0.28% “FeO _x ”, < 0.02% PbO*
Tridymite	SiO ₂		Yes	Yes	28	< 0.29% “FeO _x ”, < 0.04% PbO*
Quartz	SiO ₂		Yes	Yes	4	not analysed
Hematite	Fe ₂ O ₃		Yes	Yes	43	< 0.13% SiO ₂ , < 0.04% PbO*
Spinel	Fe ₃ O _{4+x}		Yes	Yes	5	< 0.15% SiO ₂ , < 0.02% PbO*
Melanotekite	PbO·FeO _{1.5} ·SiO ₂	PFS	Yes	Yes	34	33.9–36.4% FeO _x , 32.6–33.4% PbO
Barysilite	(9 – x)PbO·xFeO _{1+y} ·6SiO ₂	P _{9–x} F _x S ₆	No	Yes	5	39.3–40.1% SiO ₂ , 4.8–5.3% FeO _x
Melilite	2PbO·Fe ₂ O ₃ ·SiO ₂ 2PbO·FeO·2SiO ₂		Yes	No	0	**
5-lead iron silicate	5PbO·FeO _{1.5} ·SiO ₂	P ₅ FS	Yes	Yes	4	13.1–14.3% SiO ₂ , 71.6–72.9% PbO
6-lead iron silicate	6PbO·FeO _{1.5} ·SiO ₂ 6PbO·(1 + x)FeO _{1.5} ·(1 – x)SiO ₂	P ₆ FS P ₆ F _{1+x} S _{1–x}	No	Yes	3	9.8–11.8% SiO ₂ , 74.4–76.3% PbO
Magnetoplumbite	PbO·(12 – x)FeO _{1.5} , 0 < x < 2	P _{1+x} F _{12–x}	PbFe ₁₀ O ₁₆	Yes	20	Fe:Pb = 10–12 < 0.25% SiO ₂
Plumboferrite	PbO·(5 + x)FeO _{1.5} , 0 < x < 1	PF _{5+x}	PbFe ₄ O ₇	Yes	2	Fe:Pb = 5–6 < 1.5% SiO ₂
1:1 lead ferrite	PbO·(1 ± x)FeO _{1.5} , 0 < x < 0.2	PF _{1±x}	Pb ₂ Fe ₂ O ₅	No	7	51–56% PbO < 0.44% SiO ₂
Lead metasilicate, alamosite	PbO·SiO ₂	PS	Yes	Yes	9	49.7–50.1% SiO ₂ < 0.14% FeO _x
Lead orthosilicate	2PbO·SiO ₂	P ₂ S	Yes	Yes	9	33.3–33.8% SiO ₂ , < 0.45% FeO _x
11:3 lead silicate	11PbO·3SiO ₂ 11PbO·xFeO _z ·(3 – x)SiO ₂	P ₁₁ S ₃ P ₁₁ F _x S _{3–x}	Pb ₄ SiO ₆	Yes	2	< 0.8% FeO _x
5-lead silicate	5PbO·SiO ₂	P ₅ S	No	Yes	1	***
Massicot	PbO		Yes	Yes	5	< 0.01% SiO ₂ *, < 0.12% FeO _x *

All concentrations are in mol.% cations

*Within the uncertainty level

**Unstable at conditions of the present study

***Observed only in the Fe-free sample

Liquid → melanotekite (PFS) + quartz (SiO₂)
 + alamosite (PbSiO₃) at 993 ± 5 K (720 ± 5 °C),
 Liquid + melanotekite (PFS) → barysilite (P_{9–x}F_xS₆)
 + alamosite (PbSiO₃) at 993 ± 5 K (720 ± 5 °C),
 Liquid → melanotekite (PFS) + barysilite (P_{9–x}F_xS₆)
 + Pb₂SiO₄ at 988 ± 5 K (715 ± 5 °C),
 Liquid → melanotekite (PFS) + Pb₂SiO₄ (P₂S)
 + P₅FS at 953 ± 5 K (680 ± 5 °C),
 Liquid + magnetoplumbite (PF₁₂) → melanotekite (PFS)
 + P₅FS at 963 ± 10 K (690 ± 10 °C),
 Liquid + hematite (Fe₂O₃) → melanotekite (PFS)
 + magnetoplumbite (PF₁₂) at 1163
 ± 10 K (890 ± 10 °C), and
 Liquid + hematite (Fe₂O₃) → melanotekite (PFS)
 + quartz (SiO₂) at 1173 ± 10 K (900 ± 10 °C).

It also has two saddle (maximum) points on the boundary lines with alamosite (PbSiO₃) and Pb₂SiO₄:

Liquid → melanotekite (PFS)
 + alamosite (PbSiO₃) at 1018 ± 10 K (745 ± 10 °C),
 Liquid → melanotekite (PFS) + Pb₂SiO₄ at 998
 ± 5 K (725 ± 5 °C)

The barysilite (P_{9–x}F_xS₆) primary phase field has been determined less accurately compared to other primary phase fields in the present study due to experimental difficulties related to the sluggishness of achievement of equilibrium in the corresponding range. As it was observed in equilibrium with alamosite (PbSiO₃) and Pb₂SiO₄ below liquidus, barysilite is expected to be involved at least in three invariant points: Liquid + melanotekite (PFS) → barysilite (P_{9–x}F_xS₆) + alamosite (PbSiO₃), Liquid → PFS + barysilite (P_{9–x}F_xS₆) + Pb₂SiO₄ (see above), and Liquid → barysilite (P_{9–x}F_xS₆) + alamosite (PbSiO₃) + Pb₂SiO₄.

Table 2 Experimental compositions of phases in the PbO-¹⁵FeO-¹⁵SiO₂ system in air

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
<i>SiO₂ (tridymite) liquidus</i>						
1	SiO ₂ amp.	1373 (1100)	1173 (900)	0.5 + 68	Liquid	49
2	SiO ₂ amp.	1373 (1100)	1273 (1000)	0.5 + 17	Tridymite	2
3	SiO ₂ amp.	1323 (1050)	1273 (1000)	0.5 + 120	Liquid	17
4	SiO ₂ amp.	1323 (1050)	1273 (1000)	0.5 + 120	Tridymite	2
5	SiO ₂ o.	1423 (1150)	1373 (1100)	0.5 + 17	Liquid	15
6	SiO ₂ a + h.	1573 (1300)	1473 (1200)	0.5 + 5	Tridymite	2
7	Pt foil	1553 (1280)	1523 (1250)	0.5 + 2	Liquid	11
8	SiO ₂ o.	1593 (1320)	1573 (1300)	0.5 + 2	Tridymite	2
9	SiO ₂ a + h.	No	1673 (1400)	2	Liquid	44
10	SiO ₂ a + h.	No	1673 (1400)	2	Tridymite	5
11	SiO ₂ a + h.	No	1673 (1400)	2	Liquid	28
12	SiO ₂ a + h.	No	1673 (1400)	2	Tridymite	...
13	SiO ₂ a + h.	No	1673 (1400)	2	Liquid	8
<i>SiO₂ (cristobalite) liquidus</i>						
14	SiO ₂ o.	No	1743 (1470)	2	Tridymite	3
15	SiO ₂ a + h.	No	1743 (1470)	2	Liquid	46
16	SiO ₂ a + h.	No	1743 (1470)	2	Tridymite	4
					Liquid	51
					Tridymite	5
					Liquid	41
					Tridymite	5
					Liquid	29
					Tridymite	5
					Liquid	37
					Tridymite	7
					Liquid	55
					Tridymite	4
					Liquid	89
					Cristobalite	14
					Liquid	42
					Cristobalite	2
					Liquid	60
					Cristobalite	5

Table 2 continued

Exp. no.	Substrate	Premelt/precrySTALLIZATION T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
17	SiO ₂ a + h.	No	1743 (1470)	2	Liquid	8
					Cristobalite	4
18	SiO ₂ a + h.	No	1743 (1470)	2	Liquid	70
					Cristobalite	9
19	SiO ₂ a + h.	No	1743 (1470)	2	Liquid	22
					Cristobalite	1
20	Pt foil	No	1823 (1550)	2	Liquid	63
					Cristobalite	6
21	Pt foil	No	1823 (1550)	2	Liquid	36
					Cristobalite	7
22	Pt foil	No	1873 (1600)	2	Liquid	19
					Cristobalite	2
23	Pt foil	No	1873 (1600)	2	Liquid	17
					Cristobalite	4
24	Pt foil	No	1888 (1615)	1	Liquid	7
					Cristobalite	3
25	Pt foil	No	1896 (1623)	1	Liquid	18
					Cristobalite	3
26	Pt foil	No	1898 (1625)	0.5	Liquid 1	3
					Liquid 2	4
					Cristobalite	3
27	Pt foil	No	1903 (1630)	0.5	Liquid 1	6
					Liquid 2	15
					Cristobalite	4
28	Pt foil	No	1913 (1640)	0.5	Liquid 1	12
					Liquid 2	21
					Cristobalite	2
<i>Tridymite + spinel boundary</i>						
29	Pt foil	No	1673 (1400)	2	Liquid	18
					Tridymite	2
					Spinel	8
30	SiO ₂ a + h.	1728 (1455)	1703 (1430)	0.25 + 2	Liquid	24
					Tridymite	4
					Spinel	5

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
31	SiO ₂ a + h.	1728 (1455)	1713 (1440)	0.25 + 2	Liquid Tridymite Spinel	22 1 2
<i>Tridymite + Hematite boundary</i>						
32	Au foil	No	1183 (910)	96	Liquid Tridymite Hematite	23 3 5
33	Au foil	No	1198 (925)	96	Liquid Tridymite Hematite	19 2 3
34	SiO ₂ o.	1373 (1100)	1223 (950)	0.5 + 22	Liquid Tridymite Hematite	33 1 1
35	SiO ₂ o.	1323 (1050)	1273 (1000)	0.5 + 120	Liquid Tridymite Hematite	36 6 4
36	SiO ₂ o.	1423 (1150)	1323 (1050)	0.5 + 4	Liquid Tridymite Hematite	21 2 1
37	SiO ₂ o.	1473 (1200)	1423 (1150)	0.5 + 3	Liquid Tridymite Hematite	40 1 3
38	SiO ₂ o.	1543 (1270)	1523 (1250)	0.5 + 2	Liquid Tridymite Hematite	66 2 2
39	Pt foil	1633 (1360)	1623 (1350)	0.5 + 3	Liquid Tridymite Hematite	49 3 15
40	Pt foil	No	1643 (1370)	1.5	Liquid Tridymite Hematite	22 1 4
41	SiO ₂ a + h.	1693 (1420)	1663 (1390)	0.5 + 3	Liquid Tridymite Hematite	21 6 10

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
<i>Tridymite + hematite + melanotekite invariant</i>						
42	Au foil	No	1173 (900)	96	Liquid Tridymite Hematite Melanotekite PFS	33 3 2 5
<i>Quartz/tridymite + melanotekite (PFS) boundary</i>						
43	SiO ₂ o.	No	1023 (750)	168	Liquid Quartz Melanotekite PFS	17 ... 13
44	SiO ₂ o.	1093 (820)	1073 (800)	1 + 96	Liquid Quartz Melanotekite PFS	11 ... 7
45	SiO ₂ o.	No	1103 (830)	72	Liquid Quartz Melanotekite PFS	8 ... 21
46	Au foil	No	1153 (880)	96	Liquid Tridymite Melanotekite PFS	25 2 5
<i>Quartz + Alamosite (PS) + melanotekite (PFS) invariant (eutectic)</i>						
47	Au foil	1023/923 (750/650)	993 (720)	2400 + 120	Liquid Quartz Melanotekite PFS Alamosite PS	9 ... 5 7
<i>Hematite (Fe₂O₃) liquidus</i>						
48	Fe ₂ O ₃ o.	1223 (950)	1173 (900)	1 + 5	Liquid Hematite	22 5
49	Pt foil	1323 (1050)	1223 (950)	0.5 + 2	Liquid Hematite	6 4
50	Pt foil	1373 (1100)	1323 (1050)	0.5 + 2	Liquid Hematite	19 5
51	Pt foil	1493 (1220)	1423 (1150)	0.5 + 3	Liquid Hematite	18 9
52	Pt foil	1493 (1220)	1423 (1150)	0.5 + 3	Liquid	9

Table 2 continued

Exp. no.	Substrate	Premelt/precystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
53	Pt foil	No	1423 (1150)	2	Hematite	9
					Liquid	57
54	Pt foil	No	1473 (1200)	2	Hematite	9
					Liquid	54
55	Pt foil	No	1523 (1250)	2	Hematite	11
					Liquid	5
56	Pt foil	No	1523 (1250)	2	Hematite	2
					Liquid	31
57	Pt foil	No	1523 (1250)	2	Hematite	5
					Liquid	5
58	Pt foil	1543 (1270)	1523 (1250)	0.5 + 3	Hematite	2
					Liquid	12
59	Pt foil	1583 (1310)	1573 (1300)	0.5 + 3	Hematite	12
					Liquid	28
60	Pt foil	1583 (1310)	1573 (1300)	0.5 + 3	Hematite	8
					Liquid	7
61	Pt foil	1583 (1310)	1573 (1300)	0.5 + 3	Hematite	4
					Liquid	73
62	Pt foil	1583 (1310)	1573 (1300)	0.5 + 3	Hematite	8
					Liquid	10
63	Pt foil	No	1573 (1300)	1.5	Hematite	3
					Liquid	68
64	Pt foil	No	1573 (1300)	1.5	Hematite	8
					Liquid	3
65	Pt foil	No	1573 (1300)	1.5	Hematite	2
					Liquid	12
66	Pt foil	No	1613 (1340)	1.5	Hematite	2
					Liquid	27
67	Pt foil	No	1623 (1350)	1.5	Hematite	6
					Liquid	21
68	Pt foil	No	1623 (1350)	1.5	Hematite	2
					Liquid	6
69	Pt foil	No	1623 (1350)	1.5	Hematite	2
					Liquid	25

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
70	Pt foil	No	1623 (1350)	1.5	Hematite	4
					Liquid	7
71	Pt foil	No	1623 (1350)	1.5	Hematite	1
					Liquid	26
72	Pt foil	No	1623 (1350)	1.5	Hematite	4
					Liquid	14
73	Pt foil	No	1623 (1350)	1.5	Hematite	1
					Liquid	5
74	Pt foil	No	1653 (1380)	2	Hematite	1
					Liquid	35
75	Pt foil	No	1653 (1380)	2	Hematite	5
					Liquid	14
					Hematite	5
<i>Spinel (Fe₃O₄) liquidus</i>						
76	Pt foil	No	1673 (1400)	2	Liquid	6
					Spinel	4
77	Pt foil	No	1673 (1400)	2	Liquid	14
					Spinel	4
<i>Hematite (Fe₂O₃) + melanotekite (PFS) boundary</i>						
78	Fe ₂ O ₃ o.	1223 (950)	1163 (890)	0.5 + 18	Liquid	55
					Hematite	1
					Melanotekite PFS	19
<i>Hematite (Fe₂O₃) + magnetoplumbite (PF₁₂) boundary</i>						
79	Fe ₂ O ₃ o.	1303 (1030)	1273 (1000)	0.5 + 3	Liquid	19
					Hematite	1
					Magnetoplumbite	11
80	Pt foil	1403 (1130)	1373 (1100)	0.5 + 3	Liquid	20
					Hematite	14
					Magnetoplumbite	18
81	Pt foil	No	1523 (1250)	2	Liquid	13
					Hematite	2
					Magnetoplumbite	11
<i>Melanotekite (PFS) liquidus</i>						
82	Au foil	1143 (870)	963 (690)	0.5 + 72	Liquid	36

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
83	Au foil	1143 (870)	973 (700)	0.5 + 72	Melanotekite PFS Liquid	9 42
84	SiO ₂ o.	No	1023 (750)	168	Melanotekite PFS Liquid	7 12
85	SiO ₂ o.	No	1023 (750)	168	Melanotekite PFS Liquid	13 13
86	Au foil	1193 (920)	1023 (750)	0.5 + 72	Melanotekite PFS Liquid	13 37
87	Au foil	1223 (950)	1023 (750)	0.5 + 72	Melanotekite PFS Liquid	5 27
88	Au foil	1223 (950)	1023 (750)	0.5 + 72	Melanotekite PFS Liquid	8 24
89	Pt foil	1143 (870)	1023 (750)	0.5 + 18	Melanotekite PFS Liquid	8 30
90	Pt foil	1153 (880)	1073 (800)	1 + 72	Melanotekite PFS Liquid	5 30
91	SiO ₂ o.	1093 (820)	1073 (800)	1 + 96	Melanotekite PFS Liquid	17 25
92	Pt foil	1193 (920)	1073 (800)	0.5 + 48	Melanotekite PFS Liquid	5 14
93	Pt foil	1193 (920)	1073 (800)	0.5 + 48	Melanotekite PFS Liquid	6 3
94	Pt foil	1143 (870)	1073 (800)	0.5 + 18	Melanotekite PFS Liquid	6 40
95	SiO ₂ o.	No	1103 (830)	72	Melanotekite PFS Liquid	10 18
96	SiO ₂ o.	No	1103 (830)	72	Melanotekite PFS Liquid	21 18
97	Pt foil	1143 (870)	1103 (830)	0.5 + 18	Melanotekite PFS Liquid	21 15
98	Au foil	No	1133 (860)	72	Melanotekite PFS Liquid	5 41
99	Pt foil	1193 (920)	1143 (870)	0.5 + 72	Melanotekite PFS Liquid	10 51

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
100	Pt foil	1223 (950)	1143 (870)	1 + 72	Melanotekite PFS Liquid	19 43
101	Au foil	No	1153 (880)	72	Melanotekite PFS Liquid	20 27
102	Pt foil	1193 (920)	1173 (900)	0.5 + 17	Melanotekite PFS Liquid	3 21
103	Au foil	No	1173 (900)	72	Melanotekite PFS Liquid	... 27
104	Au foil	No	1188 (915)	72	Melanotekite PFS Liquid	3 26
105	Fe ₂ O ₃ o.	1223 (950)	1198 (925)	0.5 + 72	Melanotekite PFS Liquid	7 40
<i>Melanotekite (PFS) + alamosite (PS) boundary</i>						
106	Au foil	No	1003 (730)	96	Liquid Melanotekite PFS	14 ...
107	Au foil	No	1013 (740)	72	Alamosite PS Liquid	5 16
<i>Melanotekite (PFS) + P₂S boundary</i>						
108	Au foil	1023/923/983 (750/650/710)	993 (720)	1 + 2400 +96	Melanotekite PFS Alamosite PS	6 10
<i>Barysilite (P_{9-x}F_xS₆) + alamosite (PS) + P₂S eutectic</i>						
109	Au foil	1023/923/983 (750/650/710)	988 (715)	1 + 2400 +72	Liquid Barysilite P _{9-x} F _x S ₆ Alamosite PS	9 12 9
<i>P₅FS liquidus</i>						
110	Au foil	993 (720)	958 (685)	0.5 + 72	Liquid P ₅ FS	4 30 25
111	Au foil	993 (720)	973 (700)	0.5 + 72	Liquid P ₅ FS	20 10

Table 2 continued

Exp. no.	Substrate	Premelt/precrySTALLIZATION T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
<i>P₅FS + PF_{1±x} boundary</i>						
112	Au foil	1023 (750)	973 (700)	1 + 72	Liquid	2
					P ₅ FS	31
					PF _{1±x}	13
<i>P₅FS + P₂S + P₁₁S₃ invariant</i>						
113	Au foil	973 (700)	958 (685)	0.1 + 72	Liquid	5
					P ₅ FS	7
					P ₂ S	3
					P ₁₁ S ₃	20
<i>P₆FS + PF_{1±x} boundary</i>						
114	Au foil	1013 (740)	993 (720)	1 + 48	Liquid	37
					P ₆ FS	9
					PF _{1±x}	1
<i>P₆FS + PbO (massicot) boundary</i>						
115	Au foil	1023 (750)	983 (710)	0.5 + 48	Liquid	25
					P ₆ FS	25
					Massicot PbO	5
<i>Alamosite (PS) liquidus</i>						
116	Au foil	1023/923/983 (750/650/710)	998 (725)	1 + 2400 +72	Liquid	31
					Alamosite PS	12
<i>P₂S liquidus</i>						
117	Au foil	1023/923/983 (750/650/710)	1003 (730)	1 + 2400 +96	Liquid	30
					P ₂ S	12
<i>P₂S + P₁₁S₃ boundary</i>						
118	Au foil	983 (710)	966 (693)	1 + 72	Liquid	52
					P ₂ S	10
					P ₁₁ S ₃	35
<i>P₅S liquidus</i>						
119	Au foil	1013 (740)	1003 (730)	0.5 + 18	Liquid	13
					P ₅ S	10
<i>1:1 lead ferrite (PF_{1±x}, PbO·(1 ± x)FeO_{1.5}) liquidus</i>						
120	Au foil	1053 (780)	1033 (760)	1 + 72	Liquid	12
					PF _{1±x}	1
121	Au foil	1083 (810)	1073 (800)	0.5 + 18	Liquid	14

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
<i>Magnetoplumbite (PF₁₂, PbO·12FeO_{1.5}) liquidus</i>						
122	Au foil	1153 (880)	973 (700)	1 + 96	Liquid	2
123	Pt foil	1273 (1000)	1123 (850)	0.5 + 17	Magnetoplumbite	27
124	Pt foil	1273 (1000)	1173 (900)	0.5 + 18	Liquid	6
125	Pt foil	1273 (1000)	1173 (900)	0.5 + 18	Magnetoplumbite	35
126	Pt foil	1193 (920)	1173 (900)	0.5 + 3	Liquid	7
127	Pt foil	1273 (1000)	1223 (950)	1 + 1	Magnetoplumbite	43
128	Pt foil	1423 (1150)	1323 (1050)	0.2 + 3	Liquid	11
129	Pt foil	1443 (1170)	1423 (1150)	0.2 + 2	Magnetoplumbite	18
130	Pt foil	No	1473 (1200)	2	Liquid	11
131	Pt foil	1503 (1230)	1473 (1200)	0.5 + 2	Magnetoplumbite	21
132	Pt foil	1503 (1230)	1473 (1200)	0.5 + 2	Liquid	2
133	Pt foil	No	1473 (1200)	2	Magnetoplumbite	9
134	Pt foil	No	1523 (1250)	2	Magnetoplumbite	7
135	Pt foil	No	1573 (1300)	1	Liquid	39
136	Pt foil	No	1593 (1320)	1	Magnetoplumbite	5
<i>Magnetoplumbite (PF₁₂) + plumboferrite (PF_{5+x}) boundary</i>						
137	Au foil	1123 (850)	1023 (750)	1 + 48	Liquid	13

Table 2 continued

Exp. no.	Substrate	Premelt/precrySTALLIZATION T, K (°C)	Final equilibration T, K (°C)	Time, h*	Phase	No. of points
138	Pt foil	1153 (880)	1073 (800)	1 + 18	Plumboferrite Magnetoplumbite Liquid Plumboferrite Magnetoplumbite	7 26 44 17 2
<i>PF_{1±x} + massicot (PbO) boundary</i>						
139	Au foil	1043 (770)	1013 (740)	1 + 18	Liquid PF _{1±x} Massicot PbO	22 9 22
140	Au foil	1043 (770)	1023 (750)	1 + 20	Liquid PF _{1±x} PF _{1±x} Massicot PbO	61 13 13 14
<i>Massicot (PbO) liquidus</i>						
141	Au foil	No	1023 (750)	4	Liquid Massicot PbO	50 3
<i>Subsolidus</i>						
142	Au foil	No	710	168	Alamosite PS P ₂ S Barysilite P _{9-x} F _x S ₆	5 3 9
143	Au foil	No	705	120	Alamosite PS P ₂ S Barysilite P _{9-x} F _x S ₆	6 4 9
144	Au foil	No	700	144	Alamosite PS P ₂ S Barysilite P _{9-x} F _x S ₆	5 3 3
145	Au foil	No	695	2400	Alamosite PS P ₂ S Barysilite P _{9-x} F _x S ₆	9 5 15
146	Au foil	1023 (750)	720	1 + 72	P ₆ FS PF _{1±x} Massicot PbO	9 2 5

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
<i>SiO₂ (tridymite) liquidus</i>						
1	64.5 ± 0.2	5.9 ± 0.15	29.6 ± 0.18	101.0	34.6	56.5
	0.06 ± 0.06	0.04 ± 0.03	99.90 ± 0.05	98.1	0.01	99.99
2	63.2 ± 0.2	5.4 ± 0.06	31.4 ± 0.2	99.3	33.3	58.8
	0.13 ± 0.05	0.07 ± 0.03	99.80 ± 0.02	101.6	0.04	99.95
3	61.6 ± 0.15	7.5 ± 0.11	30.9 ± 0.12	101.4	32.0	57.0
	0.08 ± 0.04	0.14 ± 0.06	99.78 ± 0.1	99.9	0.02	99.92
4	61.2 ± 0.2	8.1 ± 0.13	30.7 ± 0.14	101.3	31.7	56.5
	0.08 ± 0.04	0.14 ± 0.06	99.78 ± 0.1	99.9	0.02	99.93
5	60.8 ± 0.19	5.9 ± 0.13	33.3 ± 0.16	101.3	31.1	60.5
	0.10 ± 0.06	0.06 ± 0.01	99.84 ± 0.06	100.2	0.03	99.97
6	52.3 ± 0.3	15.5 ± 0.16	32.2 ± 0.17	101.4	24.8	54.6
	n.a.	n.a.	~100	...	n.a.	~100
7	56.5 ± 0.2	6.6 ± 0.12	36.9 ± 0.16	102.1	27.3	63.5
	0.09 ± 0.02	0.09 ± 0.03	99.82 ± 0.04	99.5	0.02	99.96
8	46.7 ± 0.3	20.9 ± 0.2	32.4 ± 0.19	102.9	21.1	52.4
	0.04 ± 0.03	0.26 ± 0.04	99.70 ± 0.04	101.0	0.01	99.95
9	44.8 ± 0.2	15.2 ± 0.13	40.0 ± 0.18	101.7	19.5	62.1
	0.05 ± 0.04	0.16 ± 0.06	99.79 ± 0.09	99.5	0.01	99.98
10	42.2 ± 0.16	20.8 ± 0.14	37.0 ± 0.17	101.6	18.1	56.9
	0.05 ± 0.04	0.26 ± 0.05	99.69 ± 0.09	99.7	0.01	99.94
11	42.5 ± 0.2	20.2 ± 0.18	37.4 ± 0.18	101.6	18.2	57.5
	0.05 ± 0.04	0.26 ± 0.05	99.69 ± 0.09	99.7	0.01	99.93
12	38.8 ± 0.2	29.6 ± 0.3	31.6 ± 0.18	102.0	16.5	48.3
	0.05 ± 0.05	0.32 ± 0.05	99.63 ± 0.11	99.6	0.01	99.96
13	38.6 ± 0.3	30.6 ± 0.3	30.8 ± 0.3	102.3	16.4	47.1
	0.08 ± 0.03	0.42 ± 0.02	99.50 ± 0.03	100.0	0.02	99.88
<i>SiO₂ (cristobalite) liquidus</i>						
14	45.4 ± 0.5	4.0 ± 0.11	50.6 ± 0.4	101.8	19.2	76.1
	0.05 ± 0.03	0.05 ± 0.04	99.90 ± 0.05	99.9	0.01	99.98
15	40.9 ± 0.7	15.0 ± 0.5	44.2 ± 0.2	102.5	17.0	65.6
	0.07 ± 0.03	0.19 ± 0.01	99.72 ± 0.02	100.2	0.02	99.95
16	36.1 ± 1.4	26.7 ± 1.0	37.3 ± 0.5	103.7	14.8	54.7

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
17	0.06 ± 0.04	0.44 ± 0.03	99.50 ± 0.06	100.0	0.02	0.14
	25.9 ± 0.4	47.1 ± 0.3	27.0 ± 0.2	103.6	10.3	50.4
18	0.04 ± 0.04	0.49 ± 0.11	99.47 ± 0.11	99.7	0.01	0.05
	26.2 ± 1.0	45.9 ± 1.0	27.9 ± 1.0	103.6	10.2	51.7
19	0.04 ± 0.04	0.49 ± 0.11	99.47 ± 0.11	99.7	0.01	0.04
	11.4 ± 0.3	69.6 ± 0.5	19.0 ± 0.3	104.5	4.2	71.2
	0.00	1.0	99.0	101.7	0.00	0.26
20	31.0 ± 0.3	10.4 ± 0.15	58.7 ± 0.3	102.5	11.4	10.7
	0.01	0.17 ± 0.02	99.82 ± 0.03	100.0	0.00	0.05
21	30.2 ± 0.16	23.4 ± 0.2	46.4 ± 0.2	103.7	11.5	24.9
	0.08 ± 0.05	0.38 ± 0.05	99.54 ± 0.07	100.0	0.02	0.12
22	22.0 ± 0.3	20.5 ± 0.2	57.5 ± 0.2	102.4	7.7	19.9
	0.00	0.35 ± 0.02	99.65 ± 0.02	99.6	0.00	0.12
23	20.8 ± 0.3	35.7 ± 0.3	43.5 ± 0.4	104.0	7.5	36.0
	0.01	0.61 ± 0.10	99.38 ± 0.11	99.7	0.00	0.20
24	12.9 ± 0.9	13.7 ± 0.5	73.4 ± 1.4	101.0	4.0	12.0
	0.01	0.45 ± 0.06	99.54 ± 0.07	99.8	0.00	0.24
25	14.2 ± 0.2	18.4 ± 0.13	67.4 ± 0.2	103.3	4.6	16.5
	0.05 ± 0.04	0.48 ± 0.14	99.47 ± 0.14	100.8	0.01	0.23
26	8.3 ± 0.12	12.4 ± 0.2	79.3 ± 0.4	100.2	2.5	10.4
	16.3 ± 0.2	36.3 ± 0.5	47.4 ± 0.7	102.8	5.6	35.1
	0.03 ± 0.03	0.67 ± 0.07	99.30 ± 0.10	99.8	0.01	0.28
27	3.4 ± 0.1	8.4 ± 0.4	88.2 ± 0.4	98.6	1.0	6.7
	16.2 ± 1.0	42.7 ± 1.0	41.1 ± 0.9	102.5	5.7	42.1
	0.01 ± 0.01	0.51 ± 0.03	99.48 ± 0.03	99.9	0.01	0.24
28	2.9 ± 0.2	7.4 ± 0.4	89.7 ± 0.5	101.0	0.8	5.8
	8.7 ± 1.5	59.7 ± 1.9	31.6 ± 1.5	104.1	3.0	57.8
	0.04 ± 0.01	0.62 ± 0.01	99.34 ± 0.02	100.6	0.01	0.10
<i>Tridymite + spinel boundary</i>						
29	25.5 ± 0.4	51.0 ± 0.3	23.5 ± 0.11	102.0	10.1	56.6
	0.03 ± 0.03	0.96 ± 0.09	99.01 ± 0.09	101.4	0.01	0.29
	0.04 ± 0.03	99.87 ± 0.03	0.09 ± 0.02	102.3	0.02	99.87
30	18.4 ± 0.3	61.9 ± 0.3	19.7 ± 0.12	104.2	7.0	66.1
						26.8

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
31	0.04 ± 0.04	0.83 ± 0.10	99.13 ± 0.10	101.8	0.01	0.19
	0.06 ± 0.03	99.83 ± 0.03	0.11 ± 0.01	104.1	0.02	99.84
	12.9 ± 0.3	69.2 ± 0.2	17.9 ± 0.2	104.0	4.8	71.7
	0.00	0.61	99.39	100.8	0.00	0.12
32	0.00	99.89 ± 0.01	0.11 ± 0.01	103.3	0.00	99.85
	63.0 ± 0.17	7.5 ± 0.12	29.5 ± 0.13	101.1	33.4	11.0
	0.08 ± 0.06	0.21 ± 0.03	99.71 ± 0.04	97.9	0.02	0.10
	0.06 ± 0.06	99.88 ± 0.05	0.06 ± 0.02	100.0	0.02	99.90
33	62.6 ± 0.14	7.8 ± 0.08	29.6 ± 0.11	101.4	33.0	11.5
	0.03	0.16 ± 0.01	99.81 ± 0.01	99.2	0.01	0.09
	0.09	99.89 ± 0.02	0.02 ± 0.02	100.2	0.03	99.94
	62.1 ± 0.2	8.1 ± 0.15	29.8 ± 0.2	101.9	32.5	11.8
34	0.21	0.07	99.72	100.9	0.02	0.05
	0.02	99.96	0.02	98.9	0.01	99.97
	60.7 ± 0.3	9.0 ± 0.16	30.3 ± 0.19	101.1	31.4	12.9
	0.08 ± 0.04	0.14 ± 0.06	99.78 ± 0.1	99.9	0.02	0.04
35	0.09 ± 0.03	99.86 ± 0.02	0.05 ± 0.02	99.7	0.03	99.90
	59.0 ± 0.2	10.5 ± 0.12	30.5 ± 0.19	101.9	29.9	14.9
	0.03 ± 0.03	0.17 ± 0.06	99.80 ± 0.09	99.4	0.01	0.05
	0.03	99.96	0.01	98.3	0.01	99.97
36	55.2 ± 0.2	13.6 ± 0.18	31.2 ± 0.18	101.5	27.0	18.6
	0.05	0.18	99.77	98.8	0.01	0.04
	0.09 ± 0.07	99.85 ± 0.07	0.06 ± 0.01	99.5	0.03	99.89
	49.9 ± 0.2	18.5 ± 0.16	31.6 ± 0.14	102.1	23.2	24.1
37	0.05	0.22 ± 0.06	99.73 ± 0.06	99.1	0.01	0.04
	0.05 ± 0.03	99.91 ± 0.02	0.02 ± 0.01	99.8	0.02	99.95
	43.2 ± 0.4	26.0 ± 0.3	30.8 ± 0.2	102.7	19.1	32.1
	0.05 ± 0.03	0.50 ± 0.03	99.45 ± 0.04	99.8	0.01	0.19
38	0.04 ± 0.04	99.88 ± 0.04	0.08 ± 0.02	99.8	0.01	99.88
	36.7 ± 0.7	35.6 ± 0.6	27.7 ± 0.2	103.3	15.6	42.3
	0.01	0.36	99.63	100.2	0.00	0.02
	0.03 ± 0.02	99.88 ± 0.01	0.09 ± 0.01	100.6	0.01	99.86
39	0.05 ± 0.03	99.91 ± 0.02	0.02 ± 0.01	99.8	0.02	99.95
	43.2 ± 0.4	26.0 ± 0.3	30.8 ± 0.2	102.7	19.1	32.1
	0.05 ± 0.03	0.50 ± 0.03	99.45 ± 0.04	99.8	0.01	0.19
	0.04 ± 0.04	99.88 ± 0.04	0.08 ± 0.02	99.8	0.01	99.88
40	36.7 ± 0.7	35.6 ± 0.6	27.7 ± 0.2	103.3	15.6	42.3
	0.01	0.36	99.63	100.2	0.00	0.02
	0.03 ± 0.02	99.88 ± 0.01	0.09 ± 0.01	100.6	0.01	99.86
	0.05 ± 0.03	99.91 ± 0.02	0.02 ± 0.01	99.8	0.02	99.95

Tridymite + Hematite boundary

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
41	32.2 ± 0.4 0.04 ± 0.04 0.05 ± 0.05	44.1 ± 0.5 0.45 ± 0.06 99.85 ± 0.04	23.8 ± 0.5 99.57 ± 0.05 0.10 ± 0.01	103.6 100.3 100.2	13.4 0.01 0.02	51.2 0.03 99.85
<i>Tridymite + hematite + melanotekite invariant</i>						
42	63.5 ± 0.15 0.07 ± 0.04 0.17 ± 0.01 60.7 ± 0.2	7.1 ± 0.11 0.15 ± 0.04 99.77 ± 0.01 22.8 ± 0.13	29.4 ± 0.15 99.78 ± 0.04 0.06 16.5 ± 0.15	101.7 99.6 100.1 101.9	33.8 0.02 0.04 33.0	10.6 0.06 99.94 34.7
<i>Quartz/tridymite + melanotekite (PFS) boundary</i>						
43	69.0 ± 0.18 <i>n.a.</i> 60.9 ± 0.2 68.4 ± 0.15 <i>n.a.</i> 61.0 ± 0.17 66.9 ± 0.14 <i>n.a.</i> 60.6 ± 0.2 64.7 ± 0.16 0.06 ± 0.03 60.6 ± 0.17	1.9 ± 0.08 <i>n.a.</i> 22.7 ± 0.4 2.7 ± 0.04 <i>n.a.</i> 23.0 ± 0.13 3.5 ± 0.09 <i>n.a.</i> 22.9 ± 0.4 5.5 ± 0.07 0.12 ± 0.01 22.7 ± 0.19	29.1 ± 0.2 ~100 16.4 ± 0.3 29.0 ± 0.12 ~100 16.0 ± 0.13 29.6 ± 0.17 ~100 16.4 ± 0.3 29.8 ± 0.15 99.82 ± 0.03 16.7 ± 0.16	101.4 ... 101.5 101.2 ... 101.7 101.3 ... 101.4 101.4 99.9 101.5	38.9 <i>n.a.</i> 33.2 38.3 <i>n.a.</i> 33.3 36.8 <i>n.a.</i> 33.0 34.8 0.02 32.9	3.0 <i>n.a.</i> 34.5 4.2 <i>n.a.</i> 35.1 5.4 <i>n.a.</i> 34.9 8.3 0.05 34.4
44						
45						
46						
<i>Quartz + Alamosite (PS) + melanotekite (PFS) invariant (eutectic)</i>						
47	69.4 ± 0.4 <i>n.a.</i> 60.7 ± 0.15 78.1 ± 0.08	1.5 ± 0.07 <i>n.a.</i> 22.4 ± 0.10 0.08 ± 0.04	29.1 ± 0.4 ~100 17.0 ± 0.06 21.9 ± 0.07	99.8 ... 100.5 100.2	39.3 <i>n.a.</i> 32.9 50.0	2.4 <i>n.a.</i> 33.9 0.14
<i>Hematite (Fe₂O₃) liquidus</i>						
48	76.7 ± 0.16 0.05 ± 0.03 70.5 ± 0.4 0.11 ± 0.04 66.6 ± 0.15 0.07 ± 0.03	8.9 ± 0.11 99.93 ± 0.03 9.2 ± 0.2 99.89 ± 0.04 11.5 ± 0.10 99.91 ± 0.03	14.4 ± 0.10 0.02 ± 0.01 20.3 ± 0.18 0.00 22.0 ± 0.14 0.02 ± 0.01	100.9 99.7 101.1 97.2 101.1 97.4	49.8 0.02 41.7 0.04 37.5 0.03	16.2 99.95 15.2 99.96 18.1 99.96
49						
50						

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
51	64.6 ± 0.5	16.2 ± 0.3	19.2 ± 0.4	103.3	36.1	25.3
	0.02 ± 0.02	99.98 ± 0.02	0.00	100.9	0.01	99.99
52	59.4 ± 0.6	14.7 ± 0.18	25.9 ± 0.5	103.4	30.8	21.3
	0.08 ± 0.07	99.90 ± 0.07	0.02 ± 0.02	99.9	0.03	99.94
53	70.3 ± 0.3	20.8 ± 0.3	9.0 ± 0.08	101.9	43.7	36.1
	0.09 ± 0.06	99.89 ± 0.07	0.02 ± 0.01	99.9	0.03	99.95
54	67.3 ± 0.5	23.3 ± 0.5	9.4 ± 0.3	101.2	40.4	39.1
	0.07 ± 0.04	99.92 ± 0.04	0.01 ± 0.01	100.2	0.03	99.96
55	62.7 ± 0.2	27.1 ± 0.18	10.3 ± 0.03	100.7	35.7	43.1
	0.06 ± 0.04	99.92 ± 0.04	0.02 ± 0.02	99.4	0.02	99.95
56	62.4 ± 0.2	26.2 ± 0.2	11.4 ± 0.2	101.0	35.3	41.4
	0.06 ± 0.04	99.92 ± 0.04	0.02 ± 0.02	99.4	0.02	99.95
57	61.9 ± 0.12	25.3 ± 0.09	12.9 ± 0.16	100.8	34.6	39.5
	0.06 ± 0.04	99.92 ± 0.04	0.02 ± 0.02	99.4	0.02	99.95
58	55.2 ± 0.2	22.0 ± 0.2	22.8 ± 0.2	102.8	27.8	31.0
	0.09 ± 0.07	99.88 ± 0.07	0.03 ± 0.02	100.4	0.03	99.92
59	50.9 ± 0.3	25.0 ± 0.3	24.1 ± 0.19	102.5	24.6	33.7
	0.02 ± 0.02	99.93 ± 0.03	0.05 ± 0.02	100.1	0.01	99.93
60	47.1 ± 0.3	22.8 ± 0.2	30.2 ± 0.3	102.0	21.5	29.1
	0.02 ± 0.02	99.93 ± 0.03	0.05 ± 0.02	100.1	0.01	99.93
61	48.4 ± 0.3	28.4 ± 0.2	23.1 ± 0.3	101.6	23.0	37.7
	0.09 ± 0.07	99.83 ± 0.07	0.08 ± 0.02	98.9	0.03	99.86
62	43.7 ± 0.16	24.8 ± 0.17	31.4 ± 0.18	101.0	19.4	30.7
	0.09 ± 0.07	99.83 ± 0.07	0.08 ± 0.02	98.9	0.03	99.86
63	57.0 ± 0.5	32.2 ± 0.5	10.8 ± 0.12	102.1	30.7	48.4
	0.07 ± 0.06	99.91 ± 0.06	0.02 ± 0.01	100.7	0.03	99.95
64	58.4 ± 0.5	38.0 ± 0.5	3.6 ± 0.01	102.2	32.9	59.8
	0.08	99.91 ± 0.01	0.01 ± 0.01	100.9	0.03	99.95
65	58.5 ± 0.2	39.0 ± 0.3	2.5 ± 0.2	102.6	33.1	61.8
	0.08	99.91 ± 0.01	0.01 ± 0.01	100.9	0.03	99.95
66	52.6 ± 1.1	47.4 ± 1.1	0	101.8	28.4	71.6
	0.09 ± 0.08	99.91 ± 0.08	0	98.9	0.03	99.97
67	39.6 ± 0.3	32.2 ± 0.3	28.2 ± 0.3	103.3	17.1	39.0
						43.9

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
68	0.07 ± 0.04	99.86 ± 0.03	0.07 ± 0.02	100.4	99.88	0.10
	43.8 ± 0.2	35.7 ± 0.3	20.5 ± 0.3	102.7	45.9	33.9
69	0.07 ± 0.04	99.86 ± 0.03	0.07 ± 0.02	100.4	99.88	0.10
	45.7 ± 0.4	36.0 ± 0.3	18.3 ± 0.2	102.8	47.4	31.0
70	0.07 ± 0.04	99.86 ± 0.03	0.07 ± 0.02	100.4	99.88	0.10
	47.6 ± 0.5	39.3 ± 0.18	13.1 ± 0.4	101.8	53.8	23.0
	0.00	~ 100	0.00	...	~ 100	0.00
71	49.5 ± 0.5	41.5 ± 0.5	8.9 ± 0.12	102.7	58.7	16.2
	0.03 ± 0.02	99.94 ± 0.03	0.03 ± 0.02	100.0	99.95	0.04
72	51.3 ± 0.6	42.2 ± 0.9	6.5 ± 0.6	102.5	61.2	12.2
	0.00	~ 100	0.00	...	~ 100	0.00
73	52.3 ± 0.7	45.2 ± 0.8	2.6 ± 0.10	102.6	67.3	4.9
	0.00	~ 100	0.00	...	~ 100	0.00
74	38.7 ± 0.6	43.1 ± 0.7	18.2 ± 0.3	102.3	53.7	29.0
	0.10 ± 0.05	99.83 ± 0.05	0.07 ± 0.02	99.9	99.87	0.09
75	37.3 ± 0.4	43.0 ± 0.3	19.6 ± 0.19	101.8	52.8	30.8
	0.10 ± 0.05	99.83 ± 0.05	0.07 ± 0.02	99.9	99.87	0.09
<i>Spinel (Fe₃O₄) liquidus</i>						
76	30.2 ± 0.2	51.8 ± 0.4	18.0 ± 0.3	102.4	60.5	26.9
	0.04 ± 0.03	99.87 ± 0.03	0.09 ± 0.03	102.3	99.87	0.11
77	31.2 ± 0.4	52.1 ± 0.5	16.8 ± 0.2	102.9	61.5	25.3
	0.04 ± 0.03	99.87 ± 0.03	0.09 ± 0.03	102.3	99.87	0.11
<i>Hematite (Fe₂O₃) + melanotekite (PFS) boundary</i>						
78	78.0 ± 0.15	9.0 ± 0.13	13.0 ± 0.08	100.3	16.8	31.4
	0.03	99.94	0.03	100.7	99.95	0.04
	60.4 ± 0.2	23.0 ± 0.17	16.6 ± 0.10	101.6	34.8	32.5
<i>Hematite (Fe₂O₃) + magnetoplumbite (PF₁₂) boundary</i>						
79	75.8 ± 0.19	13.8 ± 0.2	10.5 ± 0.2	100.7	25.3	25.0
	0.02	99.96	0.02	99.2	99.96	0.03
	19.8 ± 0.3	80.2 ± 0.3	0.06 ± 0.02	100.9	91.8	0.09
80	71.6 ± 0.2	17.4 ± 0.19	10.9 ± 0.3	101.7	30.5	24.8
	0.07 ± 0.06	99.92 ± 0.06	0.01 ± 0.01	100.1	99.96	0.02
	20.3 ± 0.2	79.5 ± 0.2	0.17 ± 0.05	101.2	91.4	0.25

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
81	64.9 ± 2.0	31.1 ± 2.1	3.9 ± 0.16	100.9	39.1	52.3
	0.03 ± 0.03	99.97 ± 0.03	0.00	98.8	0.01	99.99
	20.3 ± 0.3	79.7 ± 0.2	0.03 ± 0.01	100.4	8.3	91.6
<i>Melanotekite (PFS) liquidus</i>						
82	86.7 ± 0.12	3.6 ± 0.09	9.7 ± 0.07	98.6	65.5	7.7
	60.2 ± 0.13	23.6 ± 0.19	16.2 ± 0.15	101.1	32.6	35.7
83	86.9 ± 0.09	3.6 ± 0.10	9.5 ± 0.11	98.7	66.0	7.7
	60.4 ± 0.3	23.9 ± 0.3	15.7 ± 0.10	100.7	32.8	36.4
84	70.2 ± 0.3	1.9 ± 0.10	27.9 ± 0.3	100.9	40.2	3.1
	60.9 ± 0.2	22.7 ± 0.4	16.4 ± 0.3	101.5	33.2	34.5
85	70.9 ± 0.16	1.9 ± 0.08	27.3 ± 0.15	102.3	41.0	3.0
	60.9 ± 0.2	22.7 ± 0.4	16.4 ± 0.3	101.5	33.2	34.5
86	77.1 ± 0.18	1.8 ± 0.06	21.2 ± 0.16	100.5	48.9	3.1
	60.7 ± 0.17	22.8 ± 0.2	16.5 ± 0.14	101.4	33.0	34.6
87	83.5 ± 0.11	2.7 ± 0.06	13.8 ± 0.10	100.2	59.1	5.4
	60.6 ± 0.12	23.2 ± 0.3	16.2 ± 0.3	101.9	32.9	35.2
88	83.1 ± 0.11	2.5 ± 0.07	14.4 ± 0.08	100.2	58.5	5.0
	60.6 ± 0.12	23.2 ± 0.3	16.2 ± 0.3	101.9	32.9	35.2
89	84.8 ± 0.07	5.1 ± 0.03	10.1 ± 0.07	99.2	62.3	10.5
	60.4 ± 0.16	23.8 ± 0.2	15.8 ± 0.11	101.5	32.8	36.1
90	70.8 ± 0.15	2.8 ± 0.07	26.4 ± 0.16	100.6	41.1	4.5
	60.7 ± 0.17	22.9 ± 0.14	16.4 ± 0.19	101.0	33.1	34.8
91	70.7 ± 0.3	2.6 ± 0.10	26.7 ± 0.3	101.0	40.9	4.2
	61.0 ± 0.17	23.0 ± 0.13	15.9 ± 0.13	101.7	33.3	35.1
92	76.5 ± 0.14	2.5 ± 0.10	21.0 ± 0.16	100.3	48.3	4.4
	60.6 ± 0.2	23.1 ± 0.4	16.2 ± 0.18	100.5	33.0	35.2
93	74.6 ± 0.3	2.5 ± 0.04	22.8 ± 0.26	100.3	45.7	4.4
	60.6 ± 0.2	23.1 ± 0.4	16.2 ± 0.18	100.5	33.0	35.2
94	82.9 ± 0.12	6.6 ± 0.08	10.5 ± 0.08	99.6	59.3	13.3
	60.6 ± 0.2	23.3 ± 0.3	16.1 ± 0.19	101.6	32.9	35.5
95	68.0 ± 0.3	3.5 ± 0.08	28.5 ± 0.3	101.1	38.0	5.4
	60.6 ± 0.2	23.0 ± 0.4	16.4 ± 0.3	101.4	33.0	34.9
96	69.7 ± 0.2	3.4 ± 0.09	26.9 ± 0.16	101.1	39.9	5.4

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
97	60.6 ± 0.2	23.0 ± 0.4	16.4 ± 0.3	101.4	33.0	32.2
	80.9 ± 0.14	7.9 ± 0.10	11.2 ± 0.06	99.8	56.3	28.4
98	60.3 ± 0.12	23.1 ± 0.19	16.6 ± 0.09	101.6	32.7	32.4
	66.0 ± 0.11	4.7 ± 0.07	29.2 ± 0.08	101.3	36.1	56.7
99	60.2 ± 0.2	23.3 ± 0.3	16.5 ± 0.3	101.5	32.6	32.2
	69.9 ± 0.3	4.8 ± 0.10	25.3 ± 0.2	101.2	40.3	51.9
100	61.0 ± 0.16	22.8 ± 0.2	16.3 ± 0.2	102.1	33.3	32.0
	79.2 ± 0.14	7.5 ± 0.10	13.3 ± 0.08	100.0	53.4	32.4
101	60.6 ± 0.2	22.9 ± 0.2	16.5 ± 0.11	101.5	33.0	32.3
	65.9 ± 0.19	5.5 ± 0.11	28.6 ± 0.18	101.7	36.0	55.6
102	60.5 ± 0.2	23.0 ± 0.4	16.5 ± 0.15	101.8	32.9	32.2
	74.0 ± 0.11	5.9 ± 0.09	20.1 ± 0.15	100.0	45.5	44.3
103	<i>n.a.</i>			<i>n.a.</i>		
	65.5 ± 0.16	6.4 ± 0.10	28.1 ± 0.15	101.6	35.7	54.5
104	60.5 ± 0.14	22.9 ± 0.08	16.6 ± 0.11	101.6	32.8	32.4
	64.9 ± 0.2	7.2 ± 0.09	27.9 ± 0.2	101.4	35.2	53.9
105	60.7 ± 0.17	22.8 ± 0.3	16.6 ± 0.2	102.2	33.0	32.4
	72.3 ± 0.4	7.6 ± 0.2	20.1 ± 0.3	98.7	43.6	43.6
	60.3 ± 0.4	23.2 ± 0.3	16.5 ± 0.19	100.5	32.7	32.1
<i>Melanotekite (PFS) + alamosite (PS) boundary</i>						
106	69.8 ± 0.2	1.6 ± 0.05	28.6 ± 0.2	101.0	39.8	57.6
	<i>n.a.</i>			<i>n.a.</i>		
107	78.0 ± 0.13	0.01 ± 0.01	22.0 ± 0.13	101.1	49.9	50.1
	72.1 ± 0.19	1.7 ± 0.06	26.2 ± 0.2	101.2	42.4	54.7
	60.7 ± 0.07	22.4 ± 0.07	16.8 ± 0.05	101.7	33.0	32.9
	78.1 ± 0.14	0.06 ± 0.04	21.9 ± 0.14	101.1	50.0	49.9
<i>Melanotekite (PFS) + P₂S boundary</i>						
108	83.7 ± 0.14	2.0 ± 0.06	14.3 ± 0.11	100.0	59.4	36.7
	61.1 ± 0.2	22.3 ± 0.18	16.5 ± 0.08	102.3	33.4	32.5
	87.7 ± 0.09	0.10 ± 0.07	12.2 ± 0.05	99.9	66.3	33.5
<i>Barysilit (P_{9-x}F_xS₆) + alamosite (PS) + P₂S eutectic</i>						
109	83.3 ± 0.15	0.93 ± 0.07	15.8 ± 0.12	99.6	58.3	39.9
	81.2 ± 0.11	2.8 ± 0.08	16.0 ± 0.07	100.2	55.4	39.3

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
<i>P₅FS liquidus</i>						
110	78.2 ± 0.16	0.04 ± 0.04	21.7 ± 0.16	100.7	50.2	49.7
	87.5 ± 0.07	0.17 ± 0.03	12.3 ± 0.05	100.3	65.9	33.8
	86.8 ± 0.13	4.7 ± 0.10	8.5 ± 0.05	99.0	66.2	9.9
	89.5 ± 0.16	6.2 ± 0.12	4.3 ± 0.07	99.1	72.9	14.0
111	87.7 ± 0.14	5.4 ± 0.10	6.9 ± 0.07	98.3	68.5	11.7
	88.9 ± 0.16	6.7 ± 0.14	4.4 ± 0.05	98.9	71.6	15.1
<i>P₅FS + PF_{1±x} boundary</i>						
112	87.2 ± 0.02	5.8 ± 0.01	7.03 ± 0.03	98.4	67.4	12.6
	88.9 ± 0.18	6.7 ± 0.10	4.4 ± 0.15	99.9	71.8	15.1
	74.8 ± 0.2	25.2 ± 0.2	0.02 ± 0.02	100.7	51.5	48.5
<i>P₅FS + P₂S + P₁₁S₃ invariant</i>						
113	88.5 ± 0.07	3.6 ± 0.03	7.9 ± 0.06	99.0	69.3	7.9
	89.0 ± 0.11	6.3 ± 0.10	4.8 ± 0.08	99.1	71.6	14.1
	87.6 ± 0.09	0.21 ± 0.03	12.2 ± 0.07	99.0	66.1	0.4
	92.9 ± 0.11	0.35 ± 0.09	6.8 ± 0.11	98.8	78.1	0.8
<i>P₆FS + PF_{1±x} boundary</i>						
114	88.3 ± 0.12	6.9 ± 0.10	4.8 ± 0.05	96.7	70.4	15.4
	90.1 ± 0.13	6.4 ± 0.14	3.6 ± 0.04	97.0	74.4	10.9
	75.1	24.9	0.04	98.8	51.9	48.0
<i>P₆FS + PbO (massicot) boundary</i>						
115	91.1 ± 0.12	2.9 ± 0.11	6.0 ± 0.05	97.3	75.0	6.6
	90.9 ± 0.17	5.3 ± 0.15	3.8 ± 0.05	98.5	75.9	12.3
	99.97 ± 0.02	0.03 ± 0.02	0.00	97.1	99.90	0.10
<i>Alamosite (PS) liquidus</i>						
116	82.2 ± 0.09	1.2 ± 0.05	16.5 ± 0.08	99.7	56.6	2.4
	77.9 ± 0.12	0.06 ± 0.05	22.0 ± 0.11	100.3	49.8	0.11
<i>P₂S liquidus</i>						
117	84.8 ± 0.09	2.0 ± 0.07	13.2 ± 0.09	99.1	61.3	4.0
	87.8 ± 0.13	0.12 ± 0.08	12.2 ± 0.09	99.1	66.2	0.24
<i>P₂S + P₁₁S₃ boundary</i>						
118	89.2 ± 0.13	2.9 ± 0.1	7.9 ± 0.08	99.6	70.5	6.3
	87.7 ± 0.14	0.18 ± 0.08	12.1 ± 0.13	100.2	66.3	0.4

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
<i>P₅S liquidus</i>						
119	93.1 ± 0.08	0.24 ± 0.06	6.6 ± 0.10	99.0	78.7	20.8
	93.1 ± 0.05	0	6.9 ± 0.05	97.4	78.5	21.5
	95.0 ± 0.04	0	5.0 ± 0.04	97.9	83.7	16.3
<i>1:1 lead ferrite (PF_{1±0.5} PbO·(1 ± x)FeO_{1.5}) liquidus</i>						
120	88.3 ± 0.4	8.4 ± 0.3	3.3 ± 0.10	99.9	71.2	9.8
	75.4	24.4	0.17	100.9	52.3	0.44
121	87.0 ± 0.14	10.3 ± 0.13	2.7 ± 0.04	98.8	69.1	7.9
	75.3 ± 0.15	24.6 ± 0.13	0.12 ± 0.02	100.3	52.1	0.30
<i>Magnetoplumbite (PF_{1.2} PbO·12FeO_{1.5}) liquidus</i>						
122	86.6 ± 0.19	6.1 ± 0.18	7.3 ± 0.07	100.1	66.4	20.5
	21.5 ± 1.4	78.5 ± 1.4	0.03 ± 0.02	101.5	8.9	0.05
123	80.3 ± 0.13	8.6 ± 0.11	11.0 ± 0.08	99.0	55.5	27.8
	19.8 ± 0.4	80.2 ± 0.4	0.05 ± 0.02	100.0	8.1	0.07
124	78.9 ± 0.16	10.1 ± 0.12	11.0 ± 0.07	100.0	53.7	27.2
	19.7 ± 0.8	80.3 ± 0.8	0.04 ± 0.04	101.2	8.1	0.06
125	77.5 ± 0.16	9.5 ± 0.14	13.0 ± 0.13	99.7	51.2	31.2
	19.7 ± 0.8	80.3 ± 0.8	0.04 ± 0.04	101.2	8.1	0.06
126	80.8 ± 0.19	12.6 ± 0.2	6.6 ± 0.08	100.4	57.6	17.3
	19.9 ± 0.2	80.1 ± 0.2	0.01	100.1	8.2	0.02
127	77.6 ± 0.3	11.5 ± 0.17	10.9 ± 0.3	100.0	51.9	26.6
	22.6 ± 0.2	77.3 ± 0.2	0.14 ± 0.03	99.1	9.4	0.21
128	75.8 ± 0.16	17.9 ± 0.15	6.3 ± 0.07	99.5	50.9	15.5
	20.0 ± 0.4	80.0 ± 0.4	0.05 ± 0.02	100.5	8.2	0.08
129	71.1 ± 0.2	22.9 ± 0.2	6.0 ± 0.07	100.2	45.4	13.9
	19.8 ± 0.2	80.2 ± 0.2	0.05 ± 0.02	101.0	8.1	0.07
130	67.4 ± 0.2	24.5 ± 0.2	8.1 ± 0.03	100.8	40.8	17.7
	19.2 ± 0.19	80.7 ± 0.19	0.07 ± 0.02	100.9	7.8	0.11
131	67.9 ± 0.5	28.4 ± 0.5	3.7 ± 0.0	100.1	42.3	8.3
	20.0 ± 0.3	80.0 ± 0.3	0.02 ± 0.02	100.5	8.2	0.03
132	67.9 ± 0.4	29.1 ± 0.4	3.1 ± 0.07	99.8	42.4	6.9
	20.0 ± 0.3	80.0 ± 0.3	0.02 ± 0.02	100.5	8.2	0.03
133	67.8 ± 0.3	32.2 ± 0.3	0	101.4	42.9	0

Table 2 continued

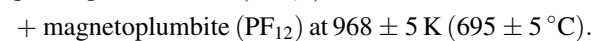
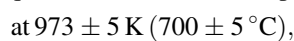
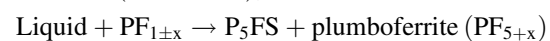
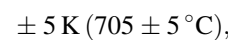
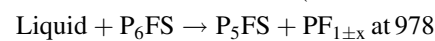
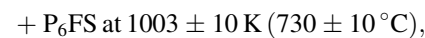
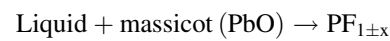
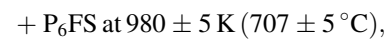
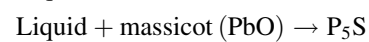
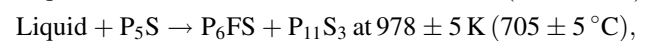
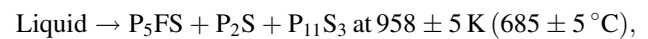
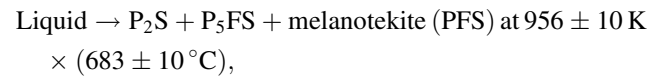
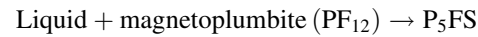
Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
134	20.8 ± 0.7	79.2 ± 0.7	0	101.9	8.6	91.4
	64.2 ± 1.0	32.7 ± 1.0	3.2 ± 0.07	100.7	38.5	54.7
135	20.3 ± 0.3	79.7 ± 0.2	0.03 ± 0.01	100.4	8.3	91.6
	58.2 ± 1.0	41.8 ± 1.0	0	102.2	33.3	66.7
136	20.1 ± 0.2	79.9 ± 0.2	0	101.2	8.3	91.7
	55.2 ± 0.7	44.8 ± 0.7	0	100.5	30.6	69.4
	20.0 ± 0.3	80.0 ± 0.3	0	101.0	8.2	91.8
<i>Magnetoplumbite (PF₁₂) + plumboferrite (PF_{5+x}) boundary</i>						
137	85.3 ± 0.19	7.6 ± 0.19	7.1 ± 0.05	99.7	64.3	15.9
	34.8 ± 2.3	64.4 ± 2.5	0.86 ± 0.15	100.5	16.0	82.5
	22.1 ± 1.2	77.9 ± 1.3	0.08 ± 0.07	99.9	9.2	90.7
138	83.9 ± 0.12	9.2 ± 0.12	6.9 ± 0.05	99.6	62.2	19.1
	31.4 ± 0.9	68.2 ± 0.9	0.39 ± 0.08	100.6	14.0	85.3
	23.7 ± 0.4	76.2 ± 0.4	0.12 ± 0.06	99.2	10.0	89.8
<i>PF_{1±x} + massicot (PbO) boundary</i>						
139	89.8 ± 0.16	7.0 ± 0.15	3.2 ± 0.05	98.9	74.2	16.2
	77.5 ± 1.2	22.5 ± 1.2	0.02 ± 0.02	100.3	55.2	44.8
	99.96 ± 0.04	0.04 ± 0.04	0.00	98.0	99.88	0.12
140	89.9 ± 0.12	8.1 ± 0.11	2.0 ± 0.04	99.7	75.1	18.8
	76.3 ± 0.3	23.7 ± 0.3	0.00	101.4	53.5	46.5
	77.9 ± 0.3	22.1 ± 0.3	0.00	101.0	55.7	44.3
	99.96 ± 0.03	0.04 ± 0.03	0.00	97.6	99.89	0.11
<i>Massicot (PbO) liquidus</i>						
141	91.9 ± 0.10	3.4 ± 0.09	4.7 ± 0.06	98.1	77.3	8.1
	99.98 ± 0.02	0.02 ± 0.02	0.00	97.5	99.94	0.06
<i>Subsolidus</i>						
142	78.0 ± 0.04	0.04 ± 0.04	21.9 ± 0.07	101.3	50.0	0.07
	87.7 ± 0.01	0.02 ± 0.02	12.3 ± 0.03	100.0	66.2	0.04
	81.0 ± 0.17	2.6 ± 0.12	16.4 ± 0.2	101.1	55.0	4.9
143	78.0 ± 0.08	0.04 ± 0.04	21.9 ± 0.07	100.5	50.0	0.07
	87.5 ± 0.12	0.15 ± 0.06	12.3 ± 0.09	98.7	65.9	0.32
	81.4 ± 0.17	2.5 ± 0.15	16.1 ± 0.07	99.8	55.5	4.9
144	78.2 ± 0.09	0.03 ± 0.03	21.7 ± 0.06	102.5	50.2	0.06

Table 2 continued

Exp. no.	Composition					
	Measured, wt.%, normalized			Total old		
	PbO	FeO _{1.5}	SiO ₂	PbO	FeO _{1.5}	SiO ₂
145	87.5 ± 0.09	0.16 ± 0.02	12.3 ± 0.09	100.7	65.9	33.7
	81.1 ± 0.06	2.8 ± 0.12	16.1 ± 0.06	102.0	55.3	39.5
	78.1 ± 0.06	0.06 ± 0.06	21.9 ± 0.08	102.0	50.0	49.8
	87.6 ± 0.09	0.16 ± 0.10	12.3 ± 0.08	99.9	66.0	33.7
	81.4 ± 0.11	2.5 ± 0.08	16.1 ± 0.05	101.1	55.6	39.6
146	90.9 ± 0.13	5.9 ± 0.13	3.2 ± 0.03	99.2	76.3	9.8
	77.0 ± 0.07	22.9 ± 0.12	0.11 ± 0.05	100.4	54.5	0.29
	99.75 ± 0.18	0.25 ± 0.18	0.00	97.7	99.31	0.00

a + h, ampoule with hole; o, open basket or crucible; n.a., not analyzed (due to small crystal size)
 *Times are listed as (pre-melt time) + (equilibration time) or (pre-melt time) + (precystallization time) + (equilibration time)
Italic values are those below the estimated uncertainty level (0.1 mol.%)
 F = FeO or FeO_{1.5} (not Fe₂O₃); hematite = Fe₂O₃; spinel (magnetite) = Fe₃O_{4+x}; melanotekite = PFS = PbO·FeO_{1.5}·SiO₂; barysilite = P_{9-x}F_xS₆ = (9 - x)PbO·xFeO_{1.5}·6SiO₂; P₅FS = 5PbO·FeO_{1.5}·SiO₂; P₆FS = 6PbO·FeO_{1.5}·SiO₂; magnetoplumbite = PF_{12-x} = PbO·(12 - x)FeO_{1.5}; 1:1 lead ferrite = PF = PF_{1±x} = PbO·(1 ± x)FeO_{1.5}

The phases P₅FS and P₆FS melt incongruently at 978 ± 5 K (705 ± 5 °C) and 998 ± 5 K (730 ± 10 °C), respectively. Their primary phase fields are surrounded by the Pb₂Fe₂O₅, plumbiferite (PF_{5+x}), magnetoplumbite (P_{1+x}F_{12-x}), melanotekite (PFS), Pb₂SiO₄ (P₂S), Pb₁₁Si₃O₁₇ (P₁₁S₃), Pb₅SiO₇ (P₅S), and massicot (PbO) primary phase fields, with the following invariant reactions being suggested:



The solidus surface of the PbO–“FeO_{1.5}”–SiO₂ system in air is presented in Fig. 4. Note that the phase relations in high-PbO corner, particularly involving the barysilite, PF_{1±x}, P₆F_{1+x}S_{1-x}, and P₁₁F_xS_{3-x} solid solutions, are not well-established in the present work.

3.1 Solid Solution Ranges in the PbO–“FeO_{1.5}”–SiO₂ System in Air

The ranges of compositions of the observed solid oxide phases in the PbO–“FeO_{1.5}”–SiO₂ system in air are listed in Table 1.

Limited solubility of iron oxide in cristobalite and tridymite is observed (Fig. 5), which reaches 0.3 mol.% “FeO_{1.5}” for equilibrium with the slags of the highest iron concentration close to the Liquid + tridymite + spinel boundary, consistent with results.^[24] Similar values are obtained for solubility of iron oxide in silica polymorphs in equilibrium with metal^[20] after correction (Fig. 5), and they can be approximated with a common trend line. Solubility of iron in quartz has not been measured, since all corresponding samples contained too small quartz crystals that gave inaccurate values. None of the silica polymorphs

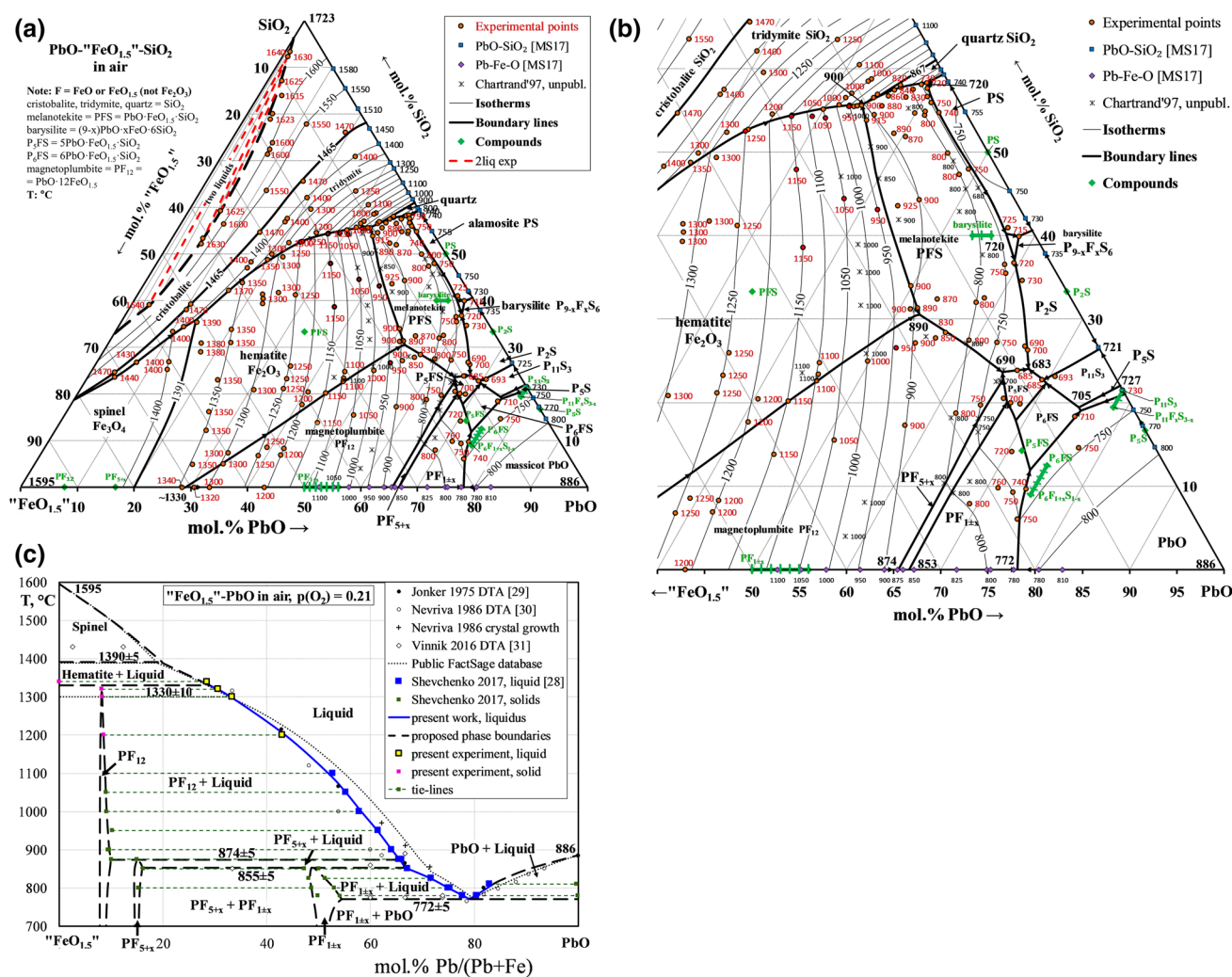


Fig. 2 Liquidus surface of the PbO- $\text{FeO}_{1.5}$ - SiO_2 system in air, $p(\text{O}_2) = 0.21$ atm. (a) complete diagram; (b) detail for Pb-rich area; (c) updated PbO- $\text{FeO}_{1.5}$ pseudobinary diagram. All temperatures

(including the points in the PbO-SiO₂ system^[26]) showed any significant solubility of PbO, < 0.03 mol.%, which is below the estimated minimum detection limit. As can be seen from comparison in Fig. 5, the differences of solubility of Fe^{2+} and Fe^{3+} in all silica polymorphs, as well as the differences between tridymite and cristobalite and the effects of temperature, are below the accuracy of the present measurements. The principal parameter that determines the solubility of $\text{FeO}_{1.5}$ in silica polymorphs is the $\text{Fe}/(\text{Fe} + \text{Pb})$ molar ratio in slag.

The solubility of silica and PbO in hematite Fe_2O_3 and spinel $\text{Fe}_3\text{O}_{4+x}$ (Fig. 6) is < 0.15 mol.% SiO_2 and < 0.04 mol.% PbO, respectively. This is much lower than

are in $^{\circ}\text{C}$. PbO-SiO₂ and PbO- $\text{FeO}_{1.5}$ binaries are based on Ref 26 and 28, respectively. Selected points^[14] given for comparison

the solubility of both SiO_2 and PbO in spinel in this system at reducing conditions.^[20] The solubility of SiO_2 shows clear trend against $\text{Si}/(\text{Pb} + \text{Si})$ molar ratio in slag, describing both hematite and spinel. However, there is no such trend for PbO solubility, indicating that the results may represent the noise from slag inclusions in hematite crystals or secondary fluorescence of PbO in slag surrounding those crystals.

Melanotekite ($\text{PbFeSiO}_{4.5}$) (Fig. 7) was found to have systematic excess of iron oxide, with the average FeO_x concentration of 34.0 mol.% instead of theoretical 33.3%. This is consistent with the results at reducing conditions.^[20] In air, the primary phase field of melanotekite

Fig. 3 Liquidus surface of the PbO–“FeO_{1.5}”–SiO₂ system in air, $p(\text{O}_2) = 0.21$ atm. (a) complete diagram; (b) detail for Pb-rich area. All temperatures are in °C and compositions in wt.%

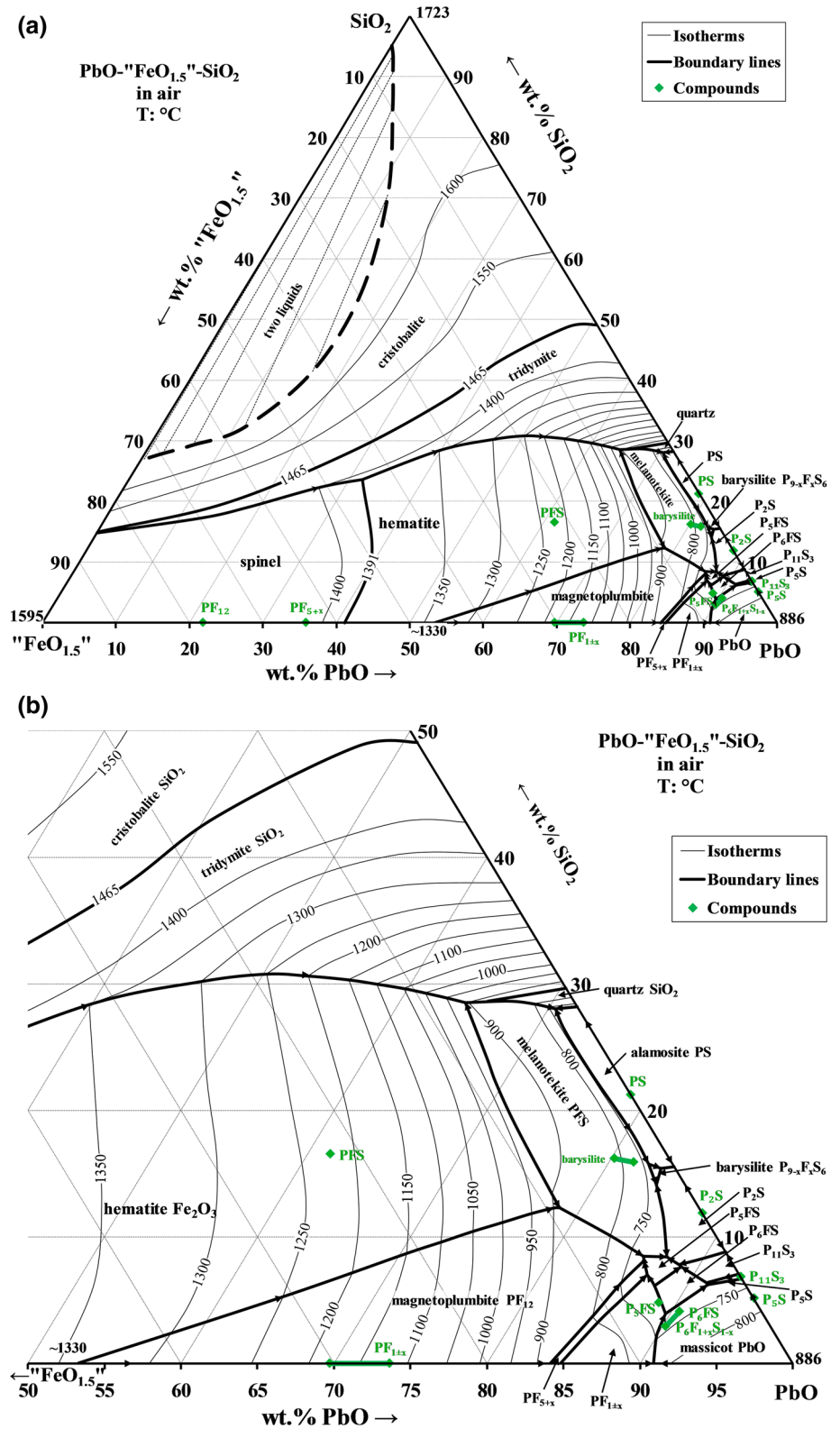
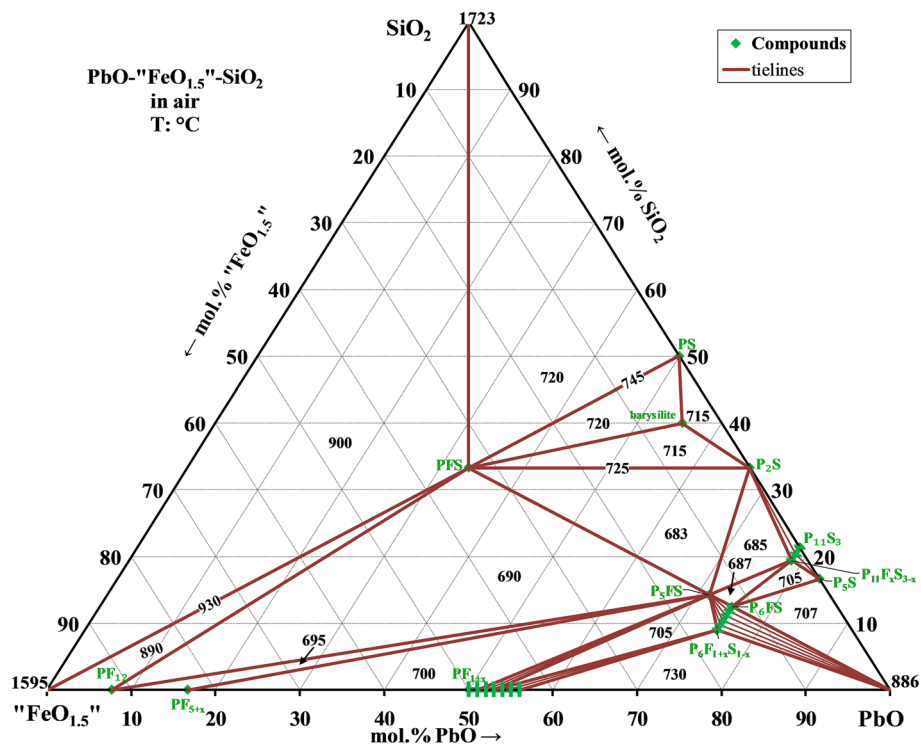


Table 3 Invariant points of the PbO–“FeO_{1.5}”–SiO₂ liquidus in air

Reaction	Estimated T, K (°C)	Liquid composition, mol.%		
		PbO	“FeO _{1.5} ”	SiO ₂
Liquid + hematite → tridymite + melanotekite	1173(900) ± 10	34	10	56
Liquid → quartz + melanotekite + PS	993(720) ± 5	40	2	58
Liquid + hematite → magnetoplumbite + melanotekite	1163(890) ± 5	52	17	31
Liquid + melanotekite + PS → barysilite	993(720) ± 5	57	3	40
Liquid → PS + P ₂ S + barysilite	988(715) ± 5	58	2	40
Liquid → melanotekite + P ₂ S + barysilite	988(715) ± 5	59	4	37
Liquid → melanotekite + magnetoplumbite + P ₅ FS	963(690) ± 10	65	12	23
Liquid + plumboferrite → + magnetoplumbite + P ₅ FS	968(695) ± 10	65	12	23
Liquid + PF _{1±x} → + plumboferrite + P ₅ FS	973(700) ± 10	67	13	20
Liquid → melanotekite + P ₂ S + P ₅ FS	956(683) ± 5	67	9	24
Liquid + P ₁₁ S ₃ → + P ₂ S + P ₅ FS	958(685) ± 5	69	8	23
Liquid + P ₆ FS → P ₁₁ S ₃ + P ₅ FS	960(687) ± 5	70	8	22
Liquid + PF _{1±x} + P ₆ FS → P ₅ FS	978(705) ± 5	69	13	18
Liquid + PF _{1±x} + PbO → P ₆ FS	1003(730) ± 10	73	15	12
Liquid + PbO → P ₆ FS + P ₅ S	980(707) ± 5	75	6	19
Liquid + P ₅ S → P ₆ FS + P ₁₁ S ₃	978(705) ± 5	75	6	19

Fig. 4 Solidus surface of the PbO–“FeO_{1.5}”–SiO₂ system in air, p(O₂) = 0.21 atm

includes much larger range of slag compositions, so it is possible to analyze the compositions of solid in more detail. As found, they can be approximated with

substitution of SiO₂ with FeO_{1.5}, keeping the Pb/(Pb + Fe + Si) ratio close to constant. The extent of substitution of SiO₂ with excess FeO_{1.5} in the ideal composition

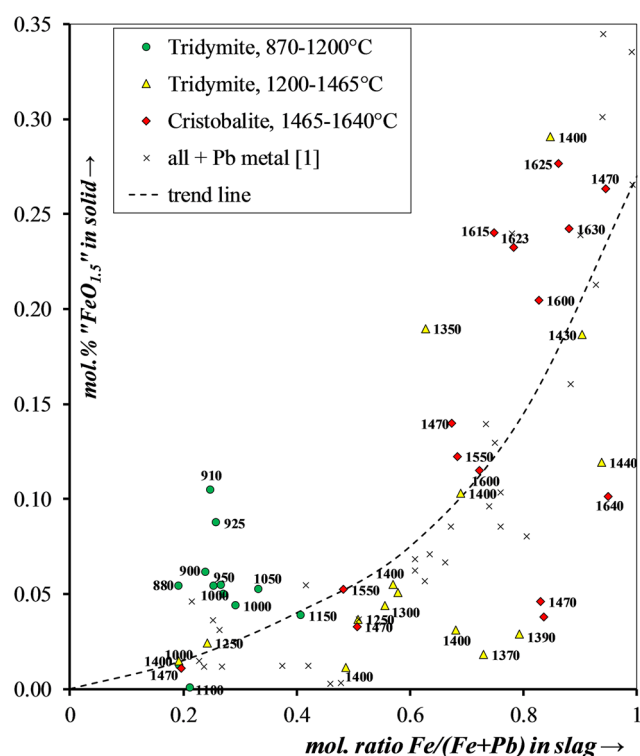


Fig. 5 Iron oxide concentrations in solid quartz, tridymite and cristobalite crystals in equilibrium with slags of the PbO - FeO_x - SiO_2 system and air, as a function of FeO_x concentration (a) and molar ratio $\text{Fe}/(\text{Fe} + \text{Pb})$ (b) in the slag. Labels indicate the experimental temperatures ($^{\circ}\text{C}$)

$\text{PbFeSiO}_{4.5}$ is the smallest (1–1.5 mol.%) when melanotekite is in equilibrium with slags close to the ratio $\text{Pb}:\text{Si} = 1$, i.e. near the $\text{PbFeSiO}_{4.5}$ - PbSiO_3 join (Fig. 7b). The largest concentration of excess iron in melanotekite (3 mol.%) is found in samples with low-silica slags, which are far from the $\text{PbFeSiO}_{4.5}$ - PbSiO_3 join.

Lead metasilicate (alamosite, $\text{PS} = \text{PbSiO}_3$) (Fig. 8a) and orthosilicate $\text{P}_2\text{S} = \text{Pb}_2\text{SiO}_4$ (Fig. 8b) are both close to stoichiometric composition, with limited solubility of FeO_x (< 0.15 and < 0.5 mol.%, respectively). Barysilite $\text{P}_{9-x}\text{F}_x\text{S}_6$ (Fig. 8c) was not studied well enough to give conclusions about its behavior: there is only one point in equilibrium with liquid, and some subsolidus points in equilibrium with PS and P_2S . The observed compositions agree with the formula $\text{Pb}_{9-x}\text{Fe}_x\text{Si}_6\text{O}_{21+y}$, $x \sim 0.7$ – 0.8 .

Only two points have been obtained for the $\text{Pb}_{11}\text{Si}_3\text{O}_{17}$ (P_{11}S_3) compound, and one (in Fe-free sample) for P_5S ; so, their deviations from stoichiometry cannot be currently analyzed in detail. Up to 0.8 mol.% FeO_x were observed in P_{11}S_3 . The Pb_5SiO_7 (P_5S) compound is unstable below 975 K (702 $^{\circ}\text{C}$).^[27]

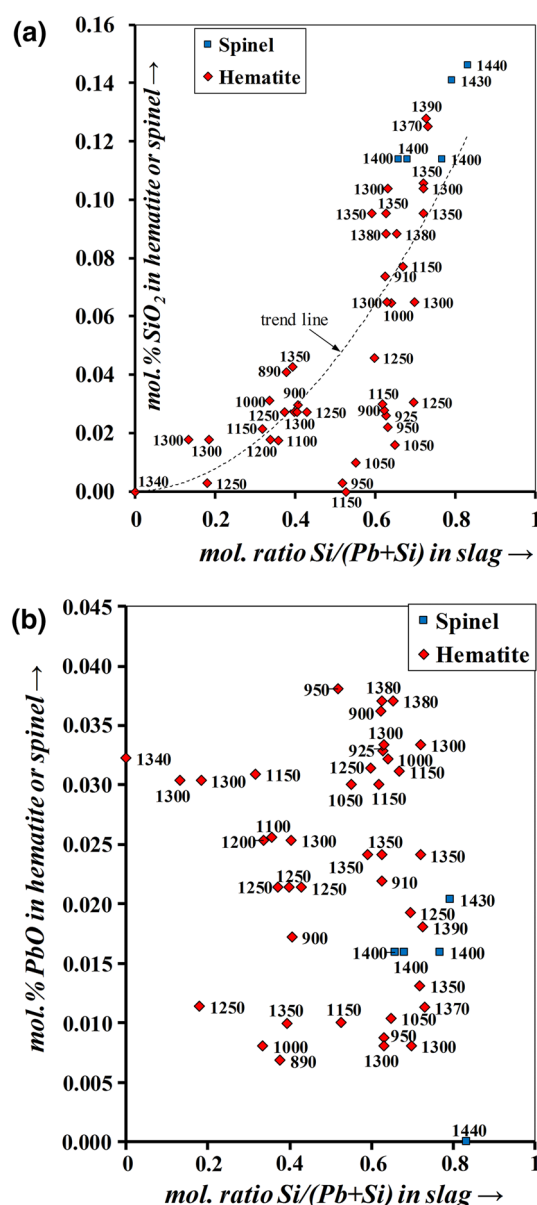


Fig. 6 SiO_2 (a) and PbO (b) concentrations in hematite and spinel crystals in equilibrium with slags of the PbO - $\text{FeO}_{1.5}$ - SiO_2 system in air, as a function of molar ratio $\text{Si}/(\text{Pb} + \text{Si})$ in the slag. Labels indicate the experimental temperatures ($^{\circ}\text{C}$)

The compositions of P_5FS ($5\text{PbO} \cdot \text{FeO}_{1.5} \cdot \text{SiO}_2$) and P_6FS ($\text{P}_6\text{F}_{1+x}\text{S}_{1-x}$, $6\text{PbO} \cdot (1+x)\text{FeO}_{1.5} \cdot (1-x)\text{SiO}_2$) measured in equilibrium with slags or at subsolidus conditions in the PbO - $\text{FeO}_{1.5}$ - SiO_2 system in air (Fig. 9), are consistent with the data from previous work at reducing conditions.^[20]

Among the three lead ferrites, magnetoplumbite (PF_{12}) and 1:1 lead ferrite ($\text{PF}_{1\pm x}$) have been studied extensively

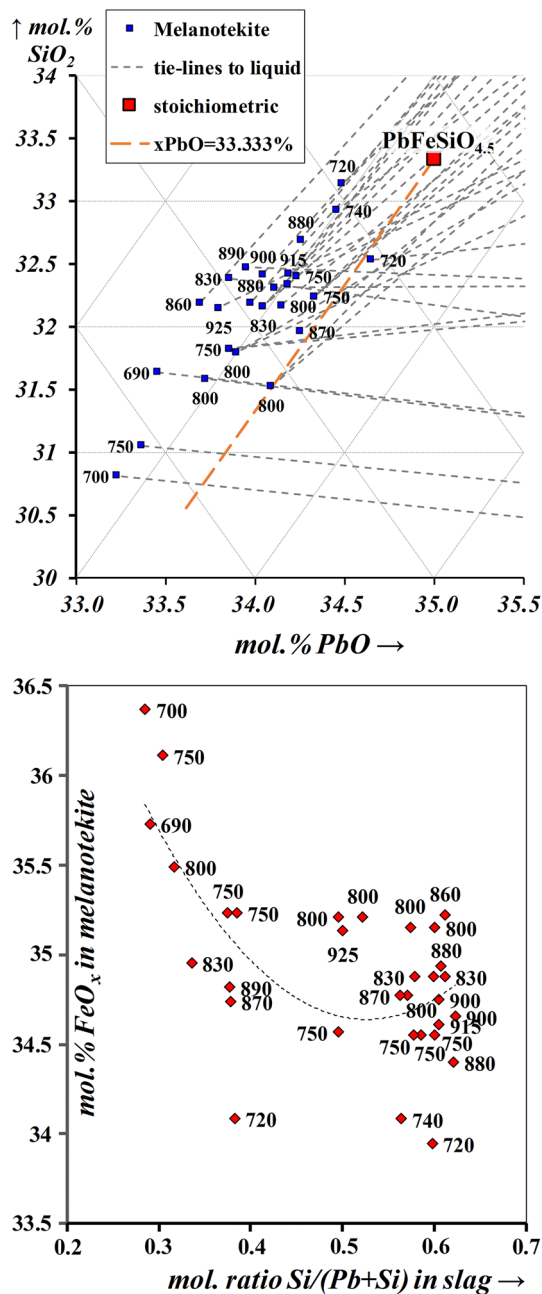


Fig. 7 Melanotekite (PFS = $\text{PbFeSiO}_{4.5}$) compositions measured on crystals in equilibrium with slags of the PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 system in air. Labels indicate the experimental temperatures ($^{\circ}\text{C}$)

in the present work. Only two points have been obtained for plumboferrite (PF_{5+x}), since its primary phase field is very narrow. Most magnetoplumbite compositions correspond to the $\text{Fe}:\text{Pb}$ molar ratio between 10 and 12 (i.e., 7.7–9.1 mol.% Pb). The points outside of this range could be contaminated with inclusions of plumboferrite or slag in magnetoplumbite crystals. The mechanism of solubility of

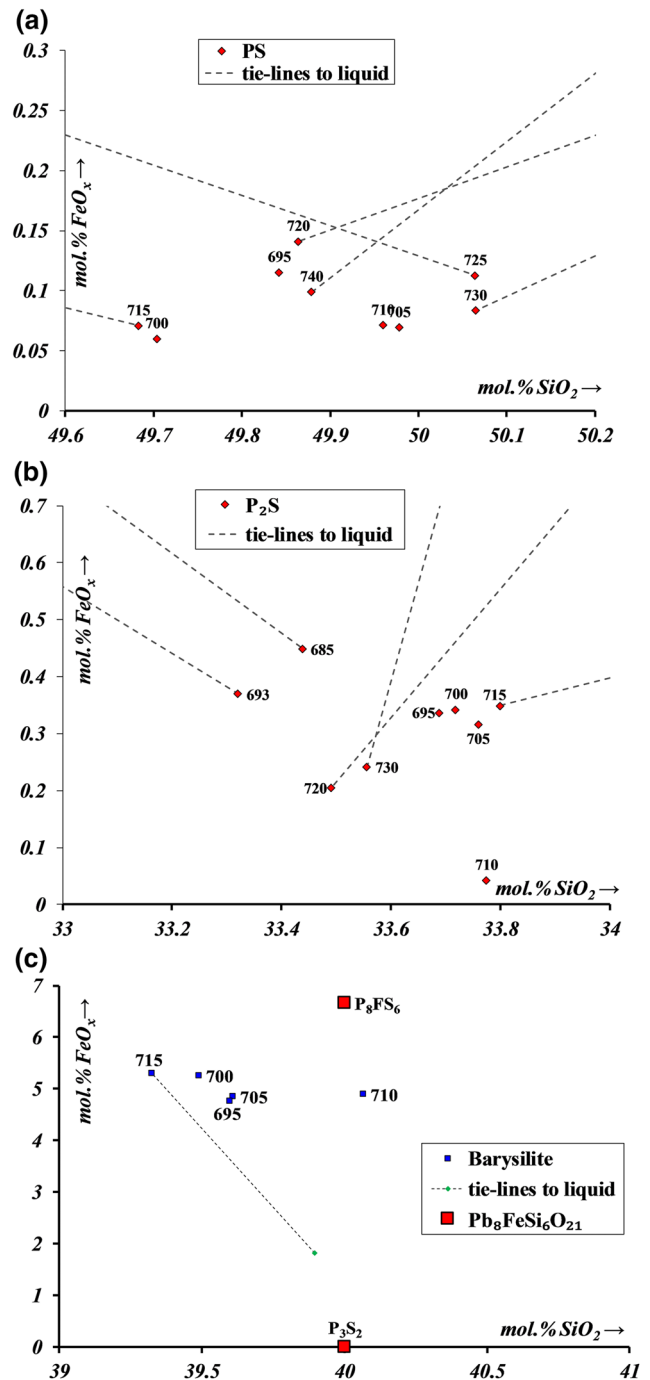


Fig. 8 PS (a), P_2S (b) and barysilite $\text{P}_{9-x}\text{F}_x\text{S}_6$ (c) compositions measured on crystals in equilibrium with slags (or at subsolidus conditions) of the PbO -“ $\text{FeO}_{1.5}$ ”- SiO_2 system in air. Labels indicate the experimental temperatures ($^{\circ}\text{C}$)

($\text{PbO} + \text{FeO}$) replacing (Fe_2O_3) in the magnetoplumbite structure described in Ref 20, 28, 34 cannot operate at oxidizing conditions due to low stability of (FeO), so the

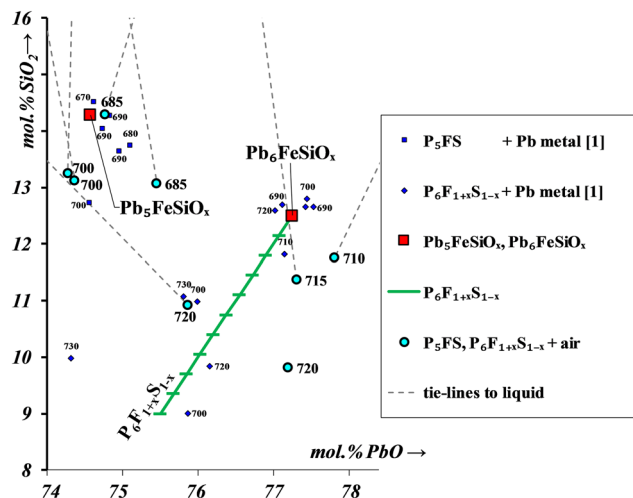


Fig. 9 P_5FS and $P_6F_{1+x}S_{1-x}$ compositions measured on crystals in equilibrium with slags (or at subsolidus conditions) of the PbO -“ $FeO_{1.5}$ ”- SiO_2 system in air, compared to the results in equilibrium with Pb metal.^[20] Labels indicate the experimental temperatures ($^{\circ}C$)

formula of magnetoplumbite can be represented as $PbO \cdot (12 - x)FeO_{1.5}$, $0 < x < 2$.

The 1:1 lead ferrite ($PF_{1 \pm x}$) has a range of solid solutions, so the points measured within the present work have 51–56 mol.% $Pb/(Pb + Fe)$. Typically, a composition of 1:1 lead ferrite is variable even within a single particle, seen as a pattern in SEM microstructures (Fig. 1f). This suggests that part of observed $PF_{1 \pm x}$ was of non-equilibrium composition, formed at initial higher temperatures during the premelt, and unable to reach the equilibrium at final temperature due to slow solid-state diffusion. All lead ferrites also showed limited solubility of SiO_2 (< 0.5 mol.%) (Fig. 10).

In summary, present experimental study has produced the first comprehensive experimental dataset for the Pb - Fe - Si - O system in air that is essential for the lead smelting and recycling industry, and provides valuable new data that can be used to optimize thermodynamic databases of these systems.

4 Conclusions

New phase equilibria information in the PbO -“ $FeO_{1.5}$ ”- SiO_2 system in air has been obtained. The studied range of temperatures covered 923–1913 K (670–1640 $^{\circ}C$), and included the equilibria of slag with one or two crystalline phases: quartz, tridymite, cristobalite (SiO_2), hematite Fe_2O_3 , spinel Fe_3O_{4+x} , melanotekite $PFS = PbO \cdot FeO_{1.5}$, barysilite $P_{9-x}F_xS_6 = (9 - x)PbO \cdot xFeO_{1+y} \cdot 6SiO_2$,

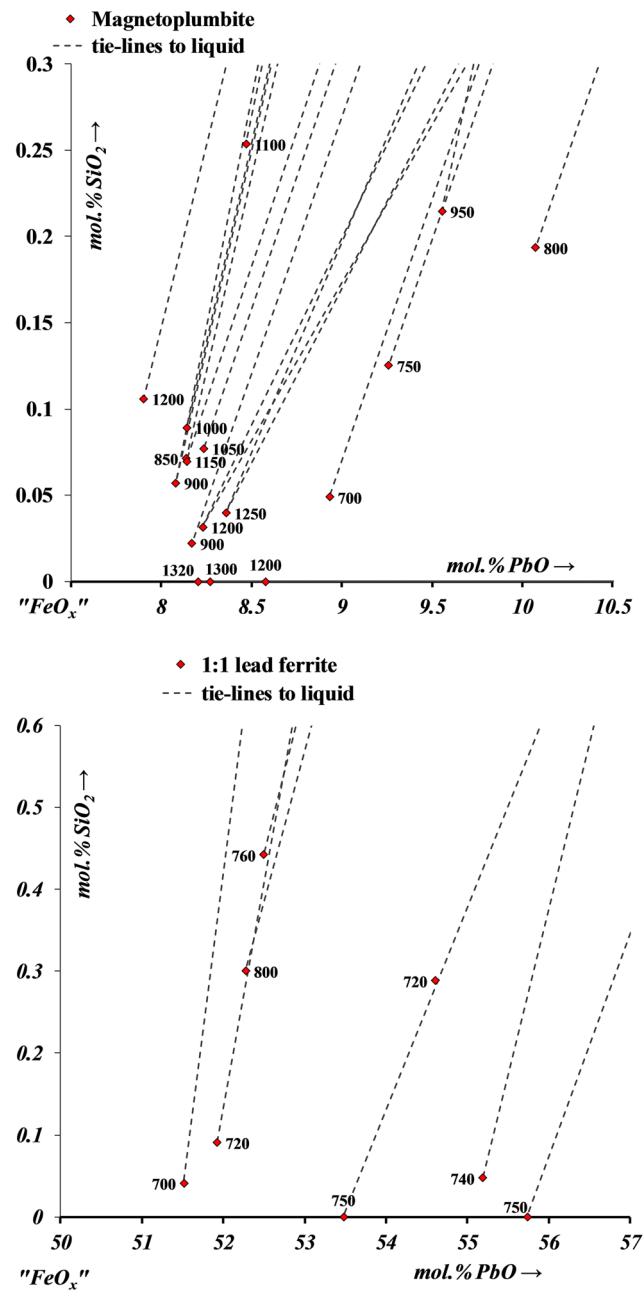


Fig. 10 Magnetoplumbite and 1:1 lead ferrite compositions in equilibrium with liquid (or at subsolidus conditions) in the PbO -“ $FeO_{1.5}$ ”- SiO_2 system in air. Labels indicate the experimental temperatures ($^{\circ}C$)

“ P_5FS ” = $5PbO \cdot FeO_{1.5} \cdot SiO_2$ and “ P_6FS ” = $6PbO \cdot (1 + x)FeO_{1.5} \cdot (1 - x)SiO_2$ ternary compounds, lead silicates (alamosite $PbSiO_3$, Pb_2SiO_4 , $Pb_{11}Si_3O_{17}$, Pb_5SiO_7), lead ferrites (magnetoplumbite $PF_{12} = PbO \cdot 12FeO_{1.5}$, plumboferrite $PbO \cdot (5 + x)FeO_{1.5}$, 1:1 lead ferrite $PbO \cdot (1 \pm x)FeO_{1.5}$), and lead oxide (PbO , massicot). This is the first

systematic study of the Pb-Fe-Si-O system in air, as a part of the multicomponent system Pb-Zn-Fe-Cu-Si-Ca-Al-Mg-O, essential for the lead smelting and recycling industries.

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